



SIH26082 Build Blueprint

Executive answer

SIH26082 expects a **Delhi-NCR-focused 72-hour air-quality forecasting system** that explicitly connects weather, atmospheric chemistry, inversion/PBL conditions and external pollution transport. The strongest feasible solution is not a purely black-box AI model and not an untested claim of live WRF-Chem; it is a **hybrid physics-ML operational system supported by a reproducible WRF-Chem hindcast and feedback experiment**.

The proposal should be framed as:

A physics-informed, uncertainty-aware Delhi-NCR air-quality early-warning and decision-support platform that forecasts PM2.5, PM10, O3 and NO2, diagnoses inversion/PBL trapping, tracks fire-derived plume influence, and demonstrates aerosol-weather feedback using WRF-Chem.

1. Official problem-statement analysis

Source status

The official SIH website was not directly retrievable in the available search results. The best accessible secondary copy identifies:

- Problem: SIH26082
- Title: Air PollutionWeather Coupled Forecasting System (Delhi NCR Focus)
- Organisation: Ministry of Earth Sciences
- Department: National Centre for Medium Range Weather Forecasting
- Category: Software
- Theme: Disaster Management
- Deadline listed by the mirror: 20 September 2026

This copy should be treated as a **secondary reproduction**, not as the official SIH portal. Before submitting the PPT, download and archive the official PS PDF from the SIH portal if it becomes accessible. ^[1]

Requirements extracted from the available PS

The accessible problem description requires a system that:

- Forecasts Delhi-NCR AQI for the next 72 hours.
- Works at high spatial resolution.
- Integrates meteorology and pollution dispersion.
- Uses WRF-Chem or a similar open-source coupled framework.
- Represents PM2.5 and ground-level ozone.
- Handles PM2.5, PM10, O3 and NOx-related chemistry.
- Includes temperature, wind and PBL height.
- Represents atmospheric inversion.
- Handles external pollution spikes such as regional stubble burning.
- Predicts stubble-burning plume dispersion.
- Shows two-way feedback between weather and chemistry.
- Provides a user-friendly real-time dashboard. ^[1]

Explicit mandatory requirements

Requirement	Practical meaning
72-hour forecasting	Produce hourly forecasts from $t + 1$ through $t + 72$
Delhi-NCR focus	Include the entire NCT Delhi and surrounding NCR districts
High resolution	Use a regional grid substantially finer than CAMS/ERA5; 1-5 km is a defensible research target
Coupled framework	Include WRF-Chem or a comparable coupled model in the scientific architecture

Requirement	Practical meaning
Pollutants	At minimum PM2.5, PM10, O3 and NOx-related variables
Weather	Temperature, wind and PBL height must be visible model variables
Inversion	Diagnose strength and duration
Stubble burning	Detect activity and model likely transport toward Delhi
Feedback	Demonstrate aerosol effects on radiation/meteorology, not only weather as an input
Dashboard	Provide spatial maps, timelines, alerts and explanations

Implicit requirements

Judges will reasonably expect:

- Data-quality control.
- Time-zone normalisation.
- Missing-data handling.
- Forecast issue-time discipline.
- Spatial interpolation or gridding.
- Historical validation.
- Multiple baselines.
- Uncertainty estimates.
- Source and data freshness labels.
- A reproducible scientific configuration.
- Clear separation between observations, hindcasts, forecasts and scenarios.
- Graceful fallback if a data source fails.

Expected outputs

Scientific outputs

- PM2.5 forecast.
- PM10 forecast.
- O3 forecast.
- NO2 or NOx forecast.
- Temperature.
- Relative humidity.
- Wind speed and direction.
- Surface pressure.
- Radiation.
- Rainfall.
- PBL height.
- Inversion score.
- Fire activity and FRP.
- Plume trajectory or influence field.

Product outputs

- Current AQI.
- 72-hour AQI timeline.
- Pollutant-specific maps.
- Station observations.
- Forecast uncertainty.
- Fire and plume maps.
- Explanation of forecast changes.
- Event replay.
- Action alerts.

Common misunderstandings

“72-hour forecast” does not mean three daily AQI values

Your output should be hourly or at least 3-hourly throughout the 72-hour period, with hourly output preferred.

“High resolution” does not mean a colourful map

A 0.25° ERA5 or 40-km CAMS field should not be described as high-resolution Delhi forecasting. Use those products as background or boundary information and produce a finer local grid.

“Coupled” does not mean “weather variables in an ML model”

An ML model using temperature, wind and PM2.5 is still a statistical model unless the chemistry affects meteorological variables or a coupled physics model is included.

“Fire detected” does not mean “fire caused the pollution”

A satellite hotspot is an observation of fire activity, not a direct measurement of emitted PM2.5 or Delhi’s source contribution.

Non-compliant claims

Avoid:

- “We predict pollution using AI and weather.”
- “FIRMS hotspots are PM2.5 emissions.”
- “Our model is two-way coupled” if no chemical feedback is demonstrated.
- “Street-level forecasting” on a 5-km grid.
- “Real-time WRF-Chem” if the output is precomputed.
- “Stubble burning caused exactly 40%” without tagged source-apportionment modelling.
- “The system replaces SAFAR or CPCB.”

What SIH expects you to build

A credible interpretation is:

A high-resolution, 72-hour Delhi-NCR pollution forecast and decision-support system that combines observations, meteorological prediction, atmospheric chemistry, inversion/PBL diagnostics, regional fire transport, uncertainty and an interactive dashboard, while demonstrating genuine weather-pollution feedback through WRF-Chem or a comparable open-source framework.

2. Delhi-NCR domain and pollution mechanism

Official NCR extent

The NCR Planning Board defines NCR as:

- Entire NCT Delhi.
- 14 Haryana districts.
- 8 Uttar Pradesh districts.
- 2 Rajasthan districts. ^[2] ^[3]

Haryana

- Faridabad.
- Gurugram.
- Nuh.
- Rohtak.
- Sonapat.
- Rewari.
- Jhajjar.
- Panipat.
- Palwal.
- Bhiwani.
- Charkhi Dadri.

- Mahendragarh.
- Jind.
- Karnal.

Uttar Pradesh

- Meerut.
- Ghaziabad.
- Gautam Budh Nagar.
- Bulandshahr.
- Baghpat.
- Hapur.
- Shamli.
- Muzaffarnagar.

Rajasthan

- Alwar.
- Bharatpur.

Delhi

- Entire NCT Delhi.

The NCRPB page is preferable to older lists because older references may omit districts added later. ^[2] ^[4]

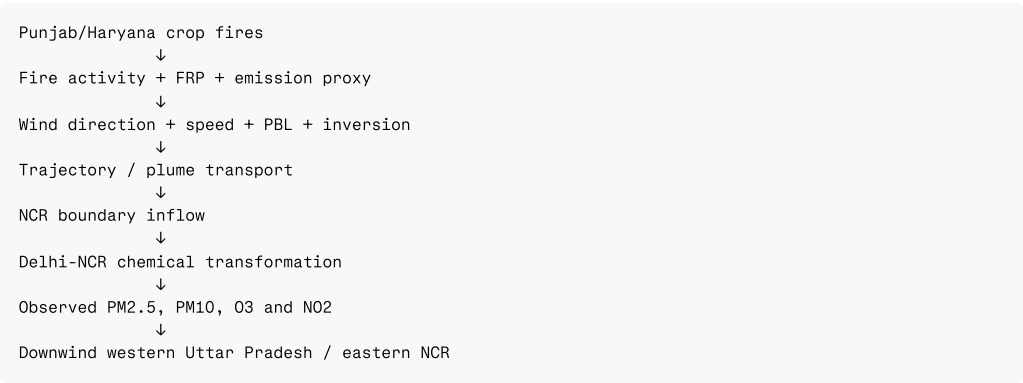
Regions outside NCR that matter

The modelling domain should extend beyond the legal NCR boundary to include:

- Punjab agricultural districts.
- Haryana agricultural districts.
- Western Uttar Pradesh.
- Northern Rajasthan.
- Chandigarh and nearby Punjab/Haryana areas.
- The wider Indo-Gangetic Plain.
- Regional dust-source areas.
- Industrial corridors and large urban centres.

Delhi is a receptor region, not an isolated air basin.

Source-to-receptor framework



Seasonal behaviour

Period	Dominant behaviour	Product implication
Winter	Weak winds, shallow PBL, inversion, fog and stagnant conditions	Strongest local accumulation and trapping alerts
Post-monsoon	Crop-residue burning, changing winds and dry conditions	Highest value for fire/plume demonstration
Monsoon	Rain and stronger mixing	Wet scavenging and lower particulate persistence
Summer	Strong convection, dust and heat	PM10/dust and ozone become more important
Pre-monsoon	Dryness, dust storms, convection and photochemistry	Useful secondary demonstration

Best demonstration period

Use **late October to mid-November**. It provides the strongest combination of:

- Fire activity.
- Regional transport.
- Low PBL.
- Inversion.
- High PM_{2.5}.
- Policy relevance.
- Clearly interpretable source-region maps.

A good primary case is **6–13 November 2017**, for which published work used 72-hour forecasting, fire emissions, MODIS AOD assimilation and aerosol-radiation effects. ^[5]

3. Scientific mechanism

Meteorology

Temperature

Temperature controls:

- Boundary-layer buoyancy.
- Reaction rates.
- Evaporation and condensation.
- Ozone chemistry.
- Stability and inversion development.

Relative humidity

Humidity affects:

- Hygroscopic particle growth.
- Visibility.
- Aqueous-phase sulfate and nitrate formation.
- Fog and cloud interactions.
- Optical depth.

Wind

Wind controls:

- Horizontal advection.
- Dilution.
- Source-region transport.
- Plume direction.
- Residence time.

Use vector components rather than raw direction:

$$u = s \sin(\theta), \quad v = s \cos(\theta)$$

with the chosen meteorological convention documented.

Atmospheric stability

Stable air reduces vertical mixing. Unstable air increases turbulent exchange and disperses surface emissions.

PBL height

PBL height is the vertical depth of the turbulent layer that mixes near-surface pollutants. A low PBL can concentrate the same emissions into a much smaller volume.

Radiation

Radiation drives:

- Surface heating.
- Convective mixing.
- Photochemical ozone formation.
- Aerosol absorption and scattering effects.

Rainfall

Rainfall removes aerosols through wet scavenging and can rapidly reduce particulate concentrations, although post-rain chemical and emission conditions determine the recovery pattern.

Surface pressure and synoptic circulation

These help identify regional weather regimes, stagnation and transport pathways.

Temperature inversion

An inversion occurs when temperature increases with height:

$$\frac{\partial T}{\partial z} > 0$$

The warmer layer above acts as a lid, suppressing vertical motion and trapping emissions near the ground.

Chemistry

PM2.5

PM2.5 contains primary and secondary material:

- Organic carbon.
- Black carbon.
- Sulfate.
- Nitrate.
- Ammonium.
- Secondary organic aerosol.
- Fine dust.

A 2024 northern-India WRF-Chem evaluation identified nitrate, dust, secondary organic aerosol and ammonium as important Delhi PM2.5 components, while also showing that models may overestimate dust and underestimate severe-event peaks. ^[6]

PM10

PM10 includes PM2.5 plus coarse particles from:

- Road dust.
- Construction.
- Soil.
- Mechanical resuspension.
- Larger combustion particles.

O3

Ozone forms through sunlight-driven reactions involving NOx and VOCs. Its relationship with PM is not monotonic: aerosol shielding, NO titration, photochemistry and vertical mixing can change the outcome.

NO, NO2 and NOx

- NO is rapidly involved in ozone titration.
- NO2 contributes to ozone chemistry and nitrate formation.
- NOx is a useful combustion-related feature, but not a unique source identifier.

SO₂, CO and VOCs

- SO₂ helps describe sulfur chemistry and industrial influence.
- CO is a useful combustion and biomass-burning tracer.
- VOCs are scientifically important for ozone but are poorly observed at dense spatial coverage.

Feedback loop

Weather to pollution

```
Weak wind + shallow PBL + inversion
      ↓
Reduced dispersion
      ↓
Higher near-surface concentration
      ↓
Higher AQI
```

Pollution to weather

```
Aerosol loading
      ↓
Scattering/absorption of radiation
      ↓
Reduced surface shortwave radiation
      ↓
Reduced surface heating
      ↓
Weaker turbulence and shallower PBL
      ↓
Further pollutant accumulation
```

Aerosol radiation interaction in WRF-Chem can reduce surface radiation, alter atmospheric dynamics and lower PBL height. Studies over the Indo-Gangetic Plain have described this feedback and its effect on aerosol concentrations. ^[7]

Genuine two-way coupling

A model is genuinely two-way coupled when:

1. Meteorology transports and chemically transforms pollutants.
2. Pollutants modify radiation, photolysis, clouds or surface energy balance.
3. The changed meteorology feeds back into transport and chemistry.

The clearest SIH evidence is:

- Run WRF-Chem with aerosol-radiation feedback off.
- Run the matched case with feedback on.
- Compare radiation, temperature, PBL height, PM_{2.5} and O₃.
- Explain the differences.

What is not genuine coupling

- ML with weather inputs.
- WRF weather followed by separate pollution mapping.
- HYSPLIT trajectories without chemistry feedback.
- Manually lowering PBL when PM_{2.5} rises.
- Displaying AOD without using it in radiation or chemical modelling.

4. Monitoring network

Current authoritative inventory

The CPCB CAAQMS station snapshot accessible for this research lists 358 stations across 31 states and Union Territories, including 39 Delhi stations in that snapshot. It provides station name, operator, coordinates and station code. CPCB states that the network changes as stations are added, relocated or retired, so this snapshot must not be treated as permanent. ^[8]

Delhi stations in the available CPCB snapshot

Station	Operator	Lat	Lon	Code
Alipur	DPCC	28.8153	77.1530	5024
Anand Vihar	DPCC	28.6476	77.3158	301
Ashok Vihar	DPCC	28.6954	77.1817	1420
Aya Nagar	IMD	28.4707	77.1099	108
Bawana	DPCC	28.7762	77.0511	1560
Burari Crossing	IMD	28.7257	77.2012	104
CRRJ Mathura Road	IMD	28.5512	77.2736	103
Chandni Chowk	IITM	28.6568	77.2272	5393
DTU	CPCB	28.7500	77.1113	118
Dr Karni Singh Shooting Range	DPCC	28.4986	77.2648	1421
Dwarka Sector 8	DPCC	28.5710	77.0719	1422
IGI Airport T3	IMD	28.5628	77.1180	106
IHBAS Dilshad Garden	CPCB	28.6812	77.3025	114
ITO	CPCB	28.6286	77.2411	117
Jahangirpuri	DPCC	28.7328	77.1706	1423
Jawaharlal Nehru Stadium	DPCC	28.5803	77.2338	1424
Lodhi Road	IITM	28.5883	77.2217	5395
Lodhi Road	IMD	28.5918	77.2273	109
Major Dhyan Chand National Stadium	DPCC	28.6113	77.2377	1425
Mandir Marg	DPCC	28.6364	77.2011	122
Mundka	DPCC	28.6847	77.0766	1561
NSIT Dwarka	CPCB	28.6091	77.0325	115
Najafgarh	DPCC	28.5702	76.9338	1427
Narela	DPCC	28.8228	77.1020	1426
Nehru Nagar	DPCC	28.5679	77.2505	1429
North Campus DU	IMD	28.6574	77.1585	105
Okhla Phase 2	DPCC	28.5308	77.2713	1428
Patparganj	DPCC	28.6238	77.2872	1431
Punjabi Bagh	DPCC	28.6740	77.1310	125
Pusa	DPCC	28.6396	77.1463	1563
Pusa	IMD	28.6396	77.1463	107
R.K. Puram	CPCB	28.5633	77.1869	124
Rohini	DPCC	28.7325	77.1199	1430
Shadipur	CPCB	28.6515	77.1473	113
Sirifort	CPCB	28.5504	77.2159	119
Sonia Vihar	DPCC	28.7105	77.2495	1432
Sri Aurobindo Marg	DPCC	28.5313	77.1902	1562
Vivek Vihar	DPCC	28.6723	77.3153	1435
Wazirpur	DPCC	28.6998	77.1655	1434

The table is a point-in-time inventory extracted from a CPCB snapshot, not a guarantee that every station is currently active or has every pollutant. Pollutant availability and current status must be queried from the live portal. ^[6]

Haryana stations of greatest importance

The snapshot includes 30 Haryana stations. For Delhi transport, prioritise:

- Bahadurgarh.
- Faridabad: New Industrial Town, Sector 11, Sector 30, Sector 16A.
- Gurugram: NISE Gwal Pahari, Sector 51, Teri Gram, Vikas Sadan.

- Manesar.
- Palwal.
- Rohtak.
- Sonipat/Murthal.
- Panipat.
- Karnal.
- Jhajjar-related locations.
- Nuh/Mandikhera.
- Rewari/Dharuhera.

Examples from the CPCB snapshot include Gurugram Sector 51 at 28.4227, 77.0667; Sonipat/Murthal at 29.0272, 77.0621; Faridabad Sector 30 at 28.4417, 77.3217; and Panipat Sector 18 at 29.4980, 76.9933. ^[8]

Uttar Pradesh stations of greatest importance

Prioritise:

- Ghaziabad: Indirapuram, Loni, Sanjay Nagar, Vasundhara.
- Noida: Sector 125, Sector 62, Sector 1, Sector 116.
- Greater Noida: Knowledge Park III and V.
- Hapur.
- Baghpat.
- Meerut.
- Bulandshahr.
- Muzaffarnagar.

Examples include Noida Sector 62 at 28.6245, 77.3577; Ghaziabad Indirapuram at 28.6462, 77.3581; and Greater Noida Knowledge Park III at 28.4727, 77.4820. ^[8]

Rajasthan stations

For NCR influence, prioritise:

- Alwar, Moti Doongri.
- Bhiwadi RIICO Industrial Area III.
- Bharatpur if present in the current live registry.

The accessible snapshot lists Alwar at 27.5548, 76.6115 and Bhiwadi at 28.1949, 76.8623. ^[9]

Outside-NCR/upwind stations

Use regional stations in:

- Punjab: Amritsar, Bathinda, Jalandhar, Khanna, Ludhiana, Mandi Gobindgarh, Patiala, Rupnagar.
- Haryana outside the immediate NCR: Ambala, Karnal, Kurukshetra, Hisar, Jind, Kaithal, Sirsa, Fatehabad.
- Western UP: Muzaffarnagar, Meerut and other available stations.
- Chandigarh.
- Rajasthan and northern India for broader transport context.

The Punjab stations in the snapshot include Ludhiana, Patiala, Khanna, Amritsar, Bathinda, Jalandhar, Mandi Gobindgarh and Rupnagar. ^[9]

Station classification

Do not infer official site type only from the station name. Derive a project classification:

- Traffic.
- Industrial.
- Residential.
- Urban background.
- Regional background.
- Mixed.

Use:

- Official metadata.
- Distance to major roads.

- Road density from OSM.
- Industrial land-use proximity.
- Building/population density.
- Manual inspection.

Recommended station table

station_id
station_name
station_code
operator
latitude
longitude
city
district
state
station_status
pollutants_available
pm25_available
pm10_available
o3_available
no_available
no2_available
nox_available
so2_available
co_available
sampling_interval
first_seen
last_seen
real_time_status
site_class
training_role
validation_role
registry_version
retrieved_at

5. Dataset catalogue

Pollution data

Dataset	Variables	Resolution/coverage	Access reality	Use
CPCB CAAQMS	PM2.5, PM10, O3, NO2, NOx, SO2, CO, NH3 where available	Station, near-hourly/hourly	Current data publicly exposed; bulk historical access must be tested	Targets and validation
CPCB AQI	AQI and pollutant sub-indices	Station/time	Official dashboard and resources	AQI verification
OGD India AQI	AQI observations	Station/time	Data.gov.in resource/API availability is resource-specific	Backup and metadata
DPCC/HSPCB/UPPCB/RSPCB	State station measurements	Station/time	Portal access varies	Regional data
OpenAQ	Aggregated station observations	Source-dependent	API and archive available; source licensing must be tracked	Gap filling and discovery
SAFAR	AQI, meteorology and forecast products	City/location-dependent	Public webpages exist; complete bulk historical access not verified	Benchmarking, not core dependency

CPCB’s real-time page and Central Control Room are the primary sources for current CAAQMS data. [\[9\]](#) [\[10\]](#) [\[11\]](#)

Meteorology

Dataset	Variables	Resolution	Access	Use
ERA5 single level	Temperature, dew point, wind, pressure, radiation, rainfall, BLH	0.25°, hourly, 1940–present	CDS registration	Historical training
ERA5 pressure level	Vertical T, RH, wind, geopotential	0.25°, 37 pressure levels	CDS registration	Inversion and stability
ERA5-Land	Land and surface variables	Approximately 0.1°, hourly	CDS	Surface energy and land context
GFS	Forecast T, wind, RH, pressure, precipitation, BLH-related fields	Forecast grid, version-dependent	NOAA	Operational weather

Dataset	Variables	Resolution	Access	Use
NCMRWF	Numerical weather products	Product-dependent	Public products exist; complete API access not verified	Indian forecast comparison
IMD	Observations and forecasts	Station/grid dependent	Access and redistribution vary	Validation if permission exists
INSAT/MOSDAC	Clouds, radiation and satellite products	Product-dependent	Registration/product-specific	Cloud and radiation features

ERA5 is available from 1940 onward and is supplied on a regular 0.25° grid; pressure-level data include vertical atmospheric variables. ^[12] ^[13]

Atmospheric composition

Dataset	Variables	Resolution	Access	Use
CAMS global	PM, O3, NO2, aerosols, gases	Approximately 40 km, 137 levels in current product descriptions	Registration/download form	Chemical background and forecast feature
CAMS reanalysis	Historical composition	Older product approximately 80 km; temporal fields product-dependent	Download	Historical background
MERRA-2	Aerosol optical depth, dust, black carbon, sulfate and meteorology	Approximately 0.5° latitude × 0.625° longitude, hourly products	NASA GES DISC	Aerosol background and evaluation
TROPOMI/Sentinel-5P	NO2, CO, O3, SO2, HCHO	Satellite pixel, product-dependent; around several kilometres	Copernicus Data Space	Regional trace-gas structure
MODIS AOD	Aerosol optical depth	Product-dependent	NASA Earthdata	Event analysis and assimilation experiments

CAMS offers global analysis and forecast products at approximately 40-km spatial scale and 137 vertical levels in current documentation. ^[14] MERRA-2 provides a long aerosol reanalysis and assimilates space-based aerosol observations. ^[15] ^[16] Sentinel-5P/TROPOMI provides daily atmospheric-constituent observations useful for air-quality analysis, but it is a column measurement with overpass and cloud limitations. ^[17] ^[18]

Fire and biomass burning

Dataset	Variables	Resolution/coverage	Access	Use
FIRMS VIIRS	Fire location, time, confidence, FRP	375-m detection pixels	API/download	Main fire feature
FIRMS MODIS	Fire location, confidence, FRP	1-km detection pixels	API/download	Historical continuity
GFED	Burned area, emissions and diurnal scaling	0.25°, 2001 onward in current product	Download	Background fire emissions
GFAS	Fire radiative power/emissions	Product-dependent	Copernicus	Fire emissions
FINN	Global fire emissions	Product-dependent	Research download	WRF-Chem input/sensitivity
State agricultural datasets	Crop residue/fire reports	Variable	Access not consistently verified	Regional validation

FIRMS gives active-fire observations, not direct emissions. FRP is reported in megawatts and should be used as a weighted activity proxy. ^[19] ^[20] ^[21]

GFED5 provides gridded monthly burned area and fire emissions from 2001 onward at 0.25° resolution, with additional scaling information for higher time resolution. ^[22] ^[23]

Emissions

Dataset	Variables	Resolution	Access/use	
EDGAR	Sectoral gases and particles	Approximately 0.1° global grid	Public products; licensing/product-specific	
HTAPv3/v3.1	SO2, NOx, CO, NMVOC, NH3, PM10, PM2.5, BC, OC	0.1°; monthly/yearly	Strong prior for WRF-Chem	
CEDS	Historical sectoral emissions	Product-dependent	Research use	
GFED/GFAS/FINN	Biomass-burning emissions	Product-dependent	Fire sensitivity	
Indian inventories	Transport, industry, residential and dust	Often regional/project-specific	Access varies	
OSM-derived proxies	Road/industrial density	Vector	ODbL obligations	Local spatial features

HTAPv3.1 includes monthly and annual gridded emissions for SO₂, NO_x, CO, NMVOC, NH₃, PM₁₀, PM_{2.5}, BC and OC at 0.1° for 2000–2020. ^[24] ^[25] ^[26]

Geography

Dataset	Use	Licensing/access
NCRPB boundaries	Official NCR domain	Government source; verify current version
Official district boundaries	Aggregation and reporting	Prefer government source
OpenStreetMap	Roads, land use, industrial features	ODbL attribution/share-alike requirements
Natural Earth	General map context	Public domain
GADM	Administrative backup	Free for academic/non-commercial use; commercial redistribution restricted
Survey of India/open government sources	Higher-authority boundaries where accessible	Terms must be checked

OpenStreetMap uses the ODbL licence and requires attribution; derivative databases may trigger share-alike obligations. ^[27] ^[28] Natural Earth data are public domain. ^[29] ^[30] GADM permits academic and non-commercial use but restricts redistribution and commercial use without permission. ^[31] ^[32]

6. Data availability reality check

Status legend

- ☐ Easily usable.
- ☐ Usable with limitations.
- ☐ Difficult.
- ☐ Practically inaccessible for the student project.

Source	Status	Reality
CPCB current observations	☐	Publicly visible, but endpoint stability and bulk access require testing
CPCB station registry	☐	Downloadable snapshot; must version by date
CPCB historical bulk data	☐/☐	Availability and extraction workflow may vary
Data.gov.in AQI	☐	API depends on resource configuration
OpenAQ	☐	Useful aggregator, but preserve source provenance and licences
ERA5	☐	Free research access with registration; not operational real-time
ERA5-Land	☐	Good historical land/surface features; not forecast
GFS	☐	Good operational fallback
CAMS	☐/☐	Strong background product; registration and product terms apply
MERRA-2	☐	Public NASA access; coarse for NCR
FIRMS NRT	☐	Practical and timely; cloud/overpass omissions remain
FIRMS archive	☐	Registration and archive workflow
MODIS AOD	☐	Cloud gaps and column-to-surface problem
TROPOMI	☐/☐	Public but satellite overpass/cloud limitations
IMD observations	☐	Institutional and redistribution restrictions may apply
NCMRWF detailed fields	☐	Public dashboards exist; full operational API not verified
SAFAR bulk history	☐	Public service exists; complete downloadable archive not verified
EDGAR/HTAP	☐/☐	Strong priors; not local real-time emissions
Indian source inventories	☐	Inconsistent public access
GADM	☐	Academic/non-commercial restrictions
Natural Earth	☐	Public domain
OSM	☐	Usable with ODbL compliance
WRF-Chem	☐ software / ☐ operation	Software is accessible; running it correctly is difficult
HYSPLIT	☐	Useful but NOAA use agreement restricts redistribution
FLEXPART	☐ software / ☐ operation	GPLv3 but demanding preprocessing
Commercial map tiles	☐	Cost and usage limits
Google Colab/Kaggle	☐ for ML / ☐ for operations	Sessions are ephemeral and unsuitable for reliable WRF cycles

7. Master dataset design

Spatial structure

Use three linked products:

1. **Station observations:** raw point observations.
2. **Operational forecast grid:** approximately 0.05° to 0.1° .
3. **Research WRF-Chem grid:** 1–3 km nested Delhi domain if compute permits.

Do not call the 0.05° grid street-level. Describe it as **regional urban-scale** or **approximately 5-km grid forecasting**.

Temporal structure

- Raw source resolution: preserved.
- Canonical modelling resolution: hourly.
- Forecast issue time: every 1–6 hours depending on operational design.
- Forecast horizon: $t + 1$ to $t + 72$.
- Historical feature windows: 72–168 hours.
- Fire aggregation windows: 6, 12, 24, 48 and 72 hours.

Feature groups

Targets

- PM2.5.
- PM10.
- O3.
- NO2.
- Optional NOx, CO and SO2.

Dynamic weather

- Temperature.
- Relative humidity.
- Dew point.
- Wind speed.
- Wind direction.
- u, v wind components.
- Surface pressure.
- Rainfall.
- Cloud cover.
- Shortwave radiation.
- PBL height.
- Mixing/ventilation coefficient.
- Vertical temperature gradients.

Dynamic chemistry

- CAMS PM2.5.
- CAMS O3.
- CAMS NO2.
- MERRA-2 AOD.
- Aerosol optical depth.
- Dust aerosol.
- CO where available.

Fire and plume

- Fire count.
- Confidence-weighted fire count.
- FRP sum.
- FRP maximum.
- Fire age.
- Distance to NCR.
- Upwind fire count.
- Upwind FRP.
- Wind-aligned FRP.
- Trajectory arrival probability.
- Source-region fire score.

Static geography

- Latitude/longitude.
- Elevation.
- Land-use class.
- Road density.
- Major-road distance.
- Industrial density.
- Population density.
- Built-up fraction.
- Vegetation fraction.
- Station type.

Temporal

- Hour of day.
- Day of week.
- Month.
- Day of year.
- Holiday indicator.
- Crop-burning-season flag.

Example schema

```
timestamp_utc
timestamp_ist
forecast_issue_time
lead_hour
grid_id
latitude
longitude
station_id
state
district
site_class

pm25_obs
pm10_obs
o3_obs
no_obs
no2_obs
nox_obs
so2_obs
co_obs

temperature_2m
relative_humidity
dewpoint_2m
surface_pressure
u10
v10
wind_speed
wind_direction
precipitation
```

```

cloud_cover
shortwave_radiation
pbl_height

temperature_925
temperature_850
temperature_700
rh_925
rh_850
wind_925
wind_850
inversion_strength
inversion_duration
inversion_layer_base
inversion_layer_top
ventilation_coefficient

cams_pm25
cams_o3
cams_no2
m2_aod
merra_dust

fire_count_6h
fire_count_24h
fire_count_48h
fire_count_72h
frp_sum_24h
frp_weighted
upwind_fire_count
upwind_frp
transport_sector
trajectory_arrival_probability

road_density
industrial_density
population_density
land_use
elevation

pm25_lag_1h
pm25_lag_3h
pm25_lag_6h
pm25_lag_12h
pm25_lag_24h
pm25_rolling_6h
pm25_rolling_24h

target_pm25_t1
target_pm25_t3
target_pm25_t6
target_pm25_t12
target_pm25_t24
target_pm25_t48
target_pm25_t72

quality_flag
source_version
data_available_at

```

Forecast formulation

For forecast issue time t_0 , create a sample:

$$X(t_0) = \{Y_{t_0-L:t_0}, W_{t_0:t_0+72}, F_{t_0-L:t_0+72}, G\}$$

where:

- Y is historical pollution.
- W is observed/future weather forecast.
- F is fire/plume/chemistry information.
- G is static geography.
- L is the historical lookback window.

Targets are:

$$Y_{t_0+h}, \quad h \in \{1, 3, 6, 12, 24, 48, 72\}$$

Prefer direct multi-horizon outputs over recursive predictions.

8. Data engineering and leakage prevention

Ingestion architecture

```
Source APIs/files
↓
Raw immutable landing zone
↓
Schema validation
↓
Unit/time normalisation
↓
Quality control
↓
Canonical hourly tables
↓
Feature generation
↓
Feature store
↓
Forecast model
```

Timestamp rules

- Store all timestamps in UTC.
- Derive IST for presentation.
- Preserve source timestamp and source timezone.
- Record publication/availability time.
- Treat ERA5 analysis as retrospective, not as a real-time forecast input.
- Use forecast issue time for operational feature availability.

Missing values

- Short gaps: limited interpolation only for auxiliary covariates.
- Pollutant targets: preserve missingness; do not create artificial targets.
- Long gaps: exclude from training or use missingness indicators.
- Weather: use model/reanalysis fallback.
- Fire: aggregate over windows and include an observation-confidence flag.

Outliers and sensor failures

Flag:

- Negative values.
- Impossible humidity.
- Flat-line sensors.
- Sudden permanent shifts.
- Repeated values.
- Long periods of zero.
- Unrealistic spikes.

Keep:

```
raw_value
clean_value
quality_flag
correction_method
```

Spatial interpolation

Use a background-plus-residual approach:

$$\hat{C}(x) = C_{\text{background}}(x) + \mathcal{I}(C_{\text{station}} - C_{\text{background,station}})$$

where $C_{\text{background}}$ comes from CAMS, WRF-Chem or another gridded product, and $\mathcal{I}$ interpolates station residuals.

Use IDW or kriging for visualisation, but evaluate whether interpolation improves independent-station validation.

Leakage controls

For every sample:

- Do not use observations later than t_0 .
- Do not use future daily averages.
- Do not compute rolling features using centred windows.
- Use only the weather forecast issued before t_0 .
- Use fire detections that would realistically have arrived by t_0 .
- Do not use revised ERA5 values as if they were available operationally.
- Split by time blocks.
- Hold out entire stations.
- Maintain an `available_at` timestamp.

9. Inversion model

Methods

Method	Scientific value	Practicality
ERA5 pressure-level lapse rate	Good regional proxy	High
WRF vertical profiles	High-resolution model diagnosis	Medium
Radiosonde profiles	Direct observational validation	Low-medium
Richardson number	Stability diagnosis using shear and buoyancy	Medium
Surface temperature/wind proxy	Operational fallback	High
PBL threshold	Trapping proxy	High

Recommended Inversion Strength Index

Use a calibrated operational index:

$$ISI = w_T Z(\Delta T_{925-1000}) + w_{T2} Z(\Delta T_{850-925}) + w_H Z(H_{\text{ref}} - H_{\text{PBL}}) + w_U Z(U_{\text{ref}} - U_{10}) + w_R Z(Ri)$$

where:

- ΔT is temperature increase with height.
- H_{PBL} is boundary-layer height.
- U_{10} is 10-m wind speed.
- Ri is a stability indicator where available.
- Z is normalisation against Delhi historical percentiles.

For a simpler version:

$$ISI = 0.35 z(\Delta T_{925-1000}) + 0.25 z(\Delta T_{850-925}) + 0.25 z(H_{\text{ref}} - H_{\text{PBL}}) + 0.15 z(U_{\text{ref}} - U_{10})$$

Do not call this a universal atmospheric standard. Call it:

Delhi-NCR pollution-trapping index

Categories

Calibrate using historical distributions:

- Low trapping: below 25th percentile.
- Moderate: 25th-75th percentile.
- High: 75th-90th percentile.
- Extreme: above 90th percentile.

Validate against:

- Radiosonde profiles.
- Low PBL periods.
- Rapid PM2.5 accumulation.
- Known smog episodes.

Dashboard

Display:

- Current index.
- Category.
- Temperature profile.
- Inversion layer.
- PBL height.
- Wind.
- Duration.
- Explanation.

Use “high trapping potential,” not unsupported universal claims such as “severe inversion.”

10. PBL height

Definition

PBL height is the approximate upper boundary of the turbulent atmospheric layer that directly mixes surface emissions.

Why it matters

For similar emissions:

- Low PBL → higher near-surface concentrations.
- High PBL → greater dilution.
- Rapid PBL collapse → evening/night pollution accumulation.
- PBL growth after sunrise → temporary dilution, although emissions and chemistry still matter.

PBL height is model-dependent and should not be presented as a direct observation unless measured using lidar, radiosonde, sodar or similar instrumentation.

Recommended data strategy

Purpose	Product
Historical training	ERA5 BLH
Operational forecast	GFS BLH or derived boundary-layer variables
Research event	WRF/WRF-Chem BLH
Validation	Radiosonde/sodar/lidar where accessible
ML feature	PBL height, change over 3/6/12 hours, PBL percentile

Include:

```
pbl_height
pbl_change_3h
pbl_change_6h
pbl_percentile
ventilation_coefficient = pbl_height × wind_speed
```

11. Stubble burning and fire-to-emission features

Fire data

Use:

- VIIRS 375-m detections for spatial detail.
- MODIS 1-km detections for longer continuity.
- Fire confidence.
- FRP.
- Acquisition time.
- Sensor and observation geometry.
- Crop calendar.

FIRMS data can be available with low latency, but exact delivery time depends on product and region. ^[19] ^[33]

Fire feature model

For source region r :

$$F_r(t) = \sum_{i \in r} FRP_i q_i e^{-\lambda \tau_i} \cdot U_i(t)$$

where:

- FRP_i is fire radiative power.
- q_i is confidence.
- τ_i is fire age.
- $U_i(t)$ indicates that the source is upwind or transport-favourable.

Use source regions:

- Punjab.
- Haryana.
- Western Uttar Pradesh.
- Rajasthan.
- Local NCR.

Emission conversion

Do not directly map FRP to PM2.5.

A more defensible emission calculation requires:

$$E_p = A_b \times FL \times CC \times EF_p$$

where:

- A_b : burned area.
- FL : fuel load.
- CC : combustion completeness.
- EF_p : pollutant-specific emission factor.

Alternatively, use FRP-based empirical emissions, but document the conversion and uncertainty.

Fire limitations

- Cloud contamination.
- Satellite overpass gaps.
- Missed small fires.
- False detections.
- Duplicate detections.
- Uncertainty in crop type.
- No direct plume height.
- No direct surface concentration.
- No direct attribution of Delhi AQI.

A 2021 analysis estimated a peak daily mean stubble-burning contribution of about 35–40% during high fire activity, while other studies and modelling systems produce different values. This disagreement is why your dashboard should show modelled influence and confidence rather than a single universal percentage. ^[34] ^[35]

12. Plume transport

Comparison

Method	Accuracy	Complexity	Compute	Real-time suitability	Recommendation
Gaussian plume	Low–medium	Low	Low	Excellent	Visual local plume only
HYSPLIT trajectory	Medium	Low–medium	Low	Good	Primary demonstration
HYSPLIT dispersion	Medium	Medium	Medium	Good	Useful operational option
FLEXPART	Medium–high	Medium–high	Medium–high	Good after automation	Strong advanced option
WRF + chemistry	High potential	High	High	Difficult	Intermediate research path

Method	Accuracy	Complexity	Compute	Real-time suitability	Recommendation
WRF-Chem	Highest physical scope	Very high	Very high	Difficult	Coupled hindcast
ML plume predictor	Variable	Medium	Low inference	Excellent	Add only after physical baseline

HYSPPLIT supports trajectories, dispersion, transport, chemical transformation and deposition, but NOAA's use agreement restricts redistribution. [\[36\]](#) [\[37\]](#) [\[38\]](#) FLEXPART is a Lagrangian transport model and its development repository is GPLv3. [\[39\]](#) [\[40\]](#) [\[41\]](#)

Recommendation

Use:

- HYSPLIT or FLEXPART for trajectory ensembles.
- Wind-aligned FRP for fire influence.
- A rasterised arrival-probability map.
- WRF-Chem for a selected historical case.
- ML for correcting concentration forecasts, not for inventing plume physics.

13. WRF-Chem feasibility

Components

A WRF-Chem simulation requires:

Meteorology

- Initial and boundary conditions.
- Terrain.
- Land use.
- Soil.
- Sea-surface temperature.
- Surface and radiation fields.
- Microphysics.
- PBL scheme.
- Land-surface model.

Chemistry

- Chemical initial and boundary conditions.
- Gas-phase mechanism.
- Aerosol scheme.
- Photolysis.
- Anthropogenic emissions.
- Biogenic emissions.
- Fire emissions.
- Dust emissions.
- Deposition.
- Plume-rise assumptions.

Model configuration

A realistic research configuration:

- D01: South Asia/northern India, 15–30 km.
- D02: Indo-Gangetic Plain, 5–10 km.
- D03: Delhi-NCR, 1–3 km if compute permits.

For the SIH demo:

- Run a coarse-to-medium test first.
- Run one historical episode.
- Perform feedback-off and feedback-on experiments.
- Save only variables required for the dashboard.

WRF-Chem schemes

The final choice depends on the installed version and validation:

- Gas chemistry: MOZART or another supported mechanism.
- Aerosol: MOSAIC or a simpler scheme for initial debugging.
- Radiation: configuration supporting aerosol-radiation interaction.
- PBL: test YSU and one alternative rather than assuming one scheme is universally best.
- Land surface: Noah or the selected supported scheme.
- Fire: FINN/GFED/GFAS or fire-derived emissions.

A 2026 India study reported a preferred WRF-Chem configuration using ERA5 boundary conditions, YSU PBL, Lin microphysics and related schemes, but this should be treated as evidence for testing—not as an automatically optimal configuration for your domain. ^[42]

WRF-Chem deliverables

- namelist.input.
- Domain map.
- WPS configuration.
- Emissions preprocessing scripts.
- Chemistry mechanism.
- Aerosol mechanism.
- Feedback flags.
- Runtime and hardware.
- Output-variable list.
- Validation plots.
- Feedback-on/off results.
- Known limitations.

Feasibility judgement

- Software access: feasible.
- One-event hindcast: feasible with Linux support.
- Multi-year high-resolution training archive: not realistic for a normal student team.
- Fully operational live WRF-Chem: risky.
- WRF-Chem-informed hybrid: feasible and defensible.

WRF is public-domain open-source code with unrestricted use according to its official repository, although the WRF name is a registered trademark. ^[43]

14. Architecture comparison

Architecture	Scientific compliance	Accuracy potential	Compute	Real-time	SIH feasibility	Demo feasibility	Verdict
A. Full operational WRF-Chem	10	8	2	3	4	5	Too risky as sole product
B. WRF-Chem hindcast + AI correction	9	9	5	7	9	9	Best overall
C. WRF + separate chemistry/plume + AI	7	8	6	8	8	8	Strong fallback
D. Physics-guided AI	7	7–8	9	9	10	9	Best MVP
E. Hybrid multi-model	9	9	6	8	9	9	Final recommendation

Final recommendation

Choose **Architecture E**, implemented as:

Scientific research layer

- WRF-Chem historical hindcast.
- Feedback-on/off comparison.
- Fire-emission sensitivity.
- PBL and inversion analysis.

Operational forecast layer

- GFS weather forecast.
- CAMS chemical background.
- CPCB observations.
- FIRMS fire features.
- HYSPLIT/FLEXPART trajectories.
- LightGBM pollutant residual correction.
- Quantile/conformal uncertainty.

Correct wording

“The operational system is a hybrid physics-informed forecast, while WRF-Chem provides the coupled research experiment and feedback evidence.”

15. ML model selection

Model comparison

Model	Input	Output	Compute	Interpretability	Suitability
Persistence	Last pollutant value	Future pollutant	Minimal	High	Required baseline
Historical mean	Seasonal/hourly averages	Future pollutant	Minimal	High	Required baseline
Random Forest	Tabular lags/weather	One or multiple horizons	Low	Medium	Baseline
XGBoost	Heterogeneous tabular features	Multi-horizon models	Low-medium	High	Strong baseline
LightGBM	Same	Multi-horizon/quantile	Low	High	Primary operational model
CatBoost	Mixed numeric/categorical data	Multi-horizon	Low-medium	Medium	Station-aware baseline
LSTM/GRU	Historical sequences	Sequence forecast	Medium	Low-medium	Deep baseline
TCN	Long sequences	Multi-horizon	Medium	Medium	Strong sequence model
CNN-LSTM	Grid/sequence	Grid forecast	Medium-high	Low	Requires dense grid
ConvLSTM	Raster sequences	Raster forecast	High	Low	Research option
Transformer/TFT	Long sequences and covariates	Multi-horizon	High	Medium	Advanced option
ST-GNN	Stations as graph	Multi-station forecast	High	Medium	Best advanced model
PINN	Physics constraints	Forecast	Very high	Theoretically high	Research only
Hybrid ensemble	Physical + ML models	Probabilistic forecast	Medium	High	Recommended

Why LightGBM first

Delhi features are heterogeneous:

- Pollutant lags.
- Meteorological forecasts.
- Fire counts and FRP.
- PBL and inversion metrics.
- CAMS fields.
- Station categories.
- Static geography.

LightGBM is efficient for this structure, offers quantile objectives and is easier to validate than a Transformer.

Research literature shows that XGBoost and ensemble approaches often perform strongly on structured air-quality data, but reported performance is highly dataset-specific and should not be copied as a guarantee for Delhi. ^[44] ^[45]

Advanced model

After the LightGBM system is validated, test:

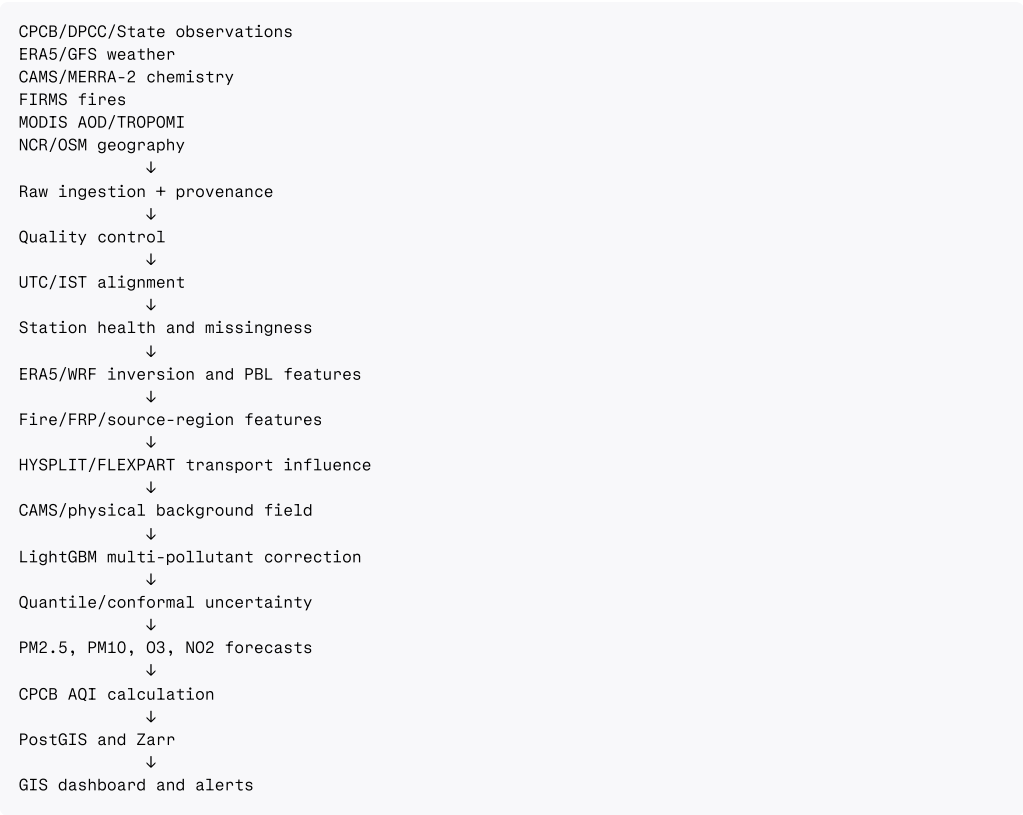
- TCN for temporal dependencies.
- ST-GNN for spatial station relationships.
- Dynamic wind-weighted graph edges.
- Ensemble of LightGBM + TCN/ST-GNN.

A wind-aware graph is preferable to a static distance-only graph:

$$A_{ij}(t) = \exp\left(-\frac{d_{ij}}{\sigma_d}\right) \cdot \max(0, \cos(\phi_{ij} - \theta_{\text{wind}}))$$

where d_{ij} is station distance, ϕ_{ij} is bearing and θ_{wind} is transport direction.

16. Forecast pipeline



Forecast equation

[
\hat Y_{i,t+h}
Y^{background}_{i,t+h}
+
f^{\theta}
\left(
Y_{t-L:t},
W_{t:t+72},
F_{t-L:t+72},
G_i,
R_{t-L:t}
\right)
]

where:

- $Y^{\text{background}}$ is CAMS, WRF-Chem or a fused physical background.
- f_θ is the ML residual model.
- R contains inversion, PBL and transport features.

17. Multiple pollutants and AQI

Pollutant-level prediction is superior

Predict:

- PM2.5.
- PM10.
- O3.
- NO2.
- Optional NOx, CO and SO2.

Then calculate AQI.

Advantages:

- Correct pollutant-specific averaging.
- Explainable AQI driver.
- Better validation.
- Support for different interventions.
- No loss of chemical information.

CPCB AQI method

India's National AQI considers eight pollutants:

- PM10.
- PM2.5.
- NO2.
- SO2.
- CO.
- O3.
- NH3.
- Pb.

The overall AQI is determined by the maximum pollutant sub-index. PM10, PM2.5, NO2, SO2, NH3 and Pb use 24-hour averages, while CO and O3 use 8-hour averages. ^[46] ^[47] ^[48]

The segmented linear formula is:

$$I_p = \frac{I_{HI} - I_{LO}}{B_{HI} - B_{LO}}(C_p - B_{LO}) + I_{LO}$$

where C_p is the averaged concentration and B and I are the relevant concentration and AQI breakpoints.

The CPCB document lists, for example:

- PM2.5: 0–30 for AQI 0–50, 31–60 for 51–100, 61–90 for 101–200, 91–120 for 201–300, 121–250 for 301–400 and 251–380 for 401–500.
- PM10: 0–50, 51–100, 101–250, 251–350, 351–430 and above 430.
- O3: 0–40, 41–80, 81–168, 169–208, 209–748 and above 748.

Use the current CPCB calculator as the versioned implementation authority. ^[46] ^[49]

18. Validation plan

Baselines

1. Persistence.
2. Historical hourly/seasonal mean.
3. Weather-only LightGBM.
4. Pollution-plus-weather LightGBM.

5. Neural AI-only model.
6. Proposed physical-background plus fire/plume/inversion ML.
7. Proposed ensemble.

Metrics

Point metrics

- MAE.
- RMSE.
- Bias.
- Correlation.
- R^2 .
- MAPE only away from low concentrations.

Skill

$$[\text{Skill}_{\text{pers}}]$$

$$1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}}$$

Probabilistic

- Prediction interval coverage.
- Mean interval width.
- CRPS.
- AQI-category probability calibration.

Operational

- Severe-event recall.
- False-alert rate.
- Time-to-warning.
- Recovery-window error.
- Data freshness.

Lead-time evaluation

Report separately at:

- 1 hour.
- 6 hours.
- 12 hours.
- 24 hours.
- 48 hours.
- 72 hours.

Regime evaluation

Separate:

- Normal days.
- High-inversion days.
- Low-PBL days.
- Severe smog.
- High-fire days.
- Rainfall/clearing events.
- Dust events.
- Ozone episodes.

Spatial evaluation

- Random temporal holdout.
- Blocked time holdout.
- Entire-station holdout.
- Event holdout.
- Season holdout.

Required plots

- Actual versus predicted PM2.5.
- Lead-time error curve.
- Forecast skill versus persistence.
- Station-wise error map.
- AQI category confusion matrix.
- Prediction interval coverage.
- Fire-event performance.
- Inversion-event performance.
- Rainfall recovery performance.
- Feedback-on/off comparison.

19. Historical Delhi episodes

6–13 November 2017

This is the strongest primary case.

Published analyses describe:

- Very high PM2.5.
- Severe haze.
- Regional transport.
- Fire and dust influence.
- Unusual meteorology.
- Low dispersion.
- Strong relevance for aerosol-radiation feedback.

A review reports PM2.5 around 710 $\mu\text{g}/\text{m}^3$ on 7 November 2017 and an AQI reported at 999 on 8 November, above the normal upper limit of the worst category. ^[50]

A MoES/IITM analysis identified interacting contributions from dust, stubble burning and local sources, with the caveat that exact percentages depend on the model configuration. ^[51]

30 October–7 November 2016

This episode is useful for:

- Severe pollution.
- Inversion and stagnation.
- Pollution accumulation.
- Regional transport.

A study of the November 2016 episode reported average PM2.5 increasing from roughly 142 $\mu\text{g}/\text{m}^3$ before the event to 563 $\mu\text{g}/\text{m}^3$ during it. ^[52]

November 2021

Useful for:

- Fire activity.
- WRF-Chem/AQWES comparison.
- Stubble-burning contribution.
- Regional transport.

A 2021 fire analysis reported maximum daily fire-related PM2.5 contributions of approximately 35–40% during peak activity, while other studies estimate different contributions. ^[24]

2–5 November 2023

Useful for:

- Recent satellite visualisation.
- MODIS and Sentinel-5P.
- Rapid PM_{2.5} increase.
- Publicly understandable event replay.

CSE reported a 68% PM_{2.5} increase within 24 hours in early November 2023, crossing approximately 313 µg/m³. ^[53]
ISRO/NICES reported increased AOD, NO₂ and CO and satellite evidence of a smoke plume across Punjab, Haryana and Delhi. ^[54]

Recommended demo set

1. Primary: 6–13 November 2017.
2. Secondary: 30 October–7 November 2016.
3. Fire-focused: early November 2021.
4. Satellite-focused: early November 2023.

Do not present reported AQI values from different station systems as directly interchangeable. Store station, averaging time, source and AQI standard.

20. Existing platforms

CPCB AQI

Strengths

- Official observations.
- National AQI method.
- Station status and current information.
- Regulatory relevance.

Weaknesses

- Primarily monitoring and communication.
- Public interfaces may not expose all scientific diagnostics.
- No single integrated public layer for inversion, PBL and fire plume attribution.

CPCB provides real-time monitoring and NCR station resources through its control-room system. ^[9] ^[10] ^[11]

SAMEER

SAMEER distributes hourly CPCB AQI updates. It is primarily an information and communication application, not a complete coupled-forecasting research environment. ^[55] ^[56]

SAFAR

SAFAR was developed by IITM with IMD weather support and historically provided location-specific forecasts for criteria pollutants including ozone, oxides of nitrogen, CO, PM_{2.5} and PM₁₀. ^[57] ^[58] ^[59]

Delhi Air Quality Early Warning System

The Delhi early-warning system is a major prior art. It was developed by IITM, IMD, NCMRWF and collaborators, and provides forecasts up to 72 hours. ^[60] ^[61] ^[62]

Therefore, your proposal must not claim that it is the first system to forecast Delhi air quality for 72 hours.

CAMS

CAMS provides global analysis and forecast products for many chemical species, but its approximately 40-km global resolution is insufficient as a standalone hyperlocal NCR solution. ^[14]

HYSPLIT/FLEXPART

These are transport and dispersion tools, not automatically two-way coupled weather–chemistry systems.

IQAir, WAQI and OpenAQ

These are useful for public communication, aggregation and data discovery, but generally do not expose the full combination of:

- WRF-Chem feedback.
- Inversion diagnostics.
- PBL state.
- Fire-to-NCR plume transport.
- Forecast uncertainty.
- Pollutant-level explainability.

Opportunity gap

Position your system as:

A transparent, local decision-support layer that connects official observations and forecasts with physical explanations, regional fire transport, inversion/PBL diagnosis and uncertainty.

Do not position it as a replacement for established government systems.

21. Competitor/prior-art matrix

“Unknown” means the feature was not verified from the accessible source and must not be claimed.

System	Weather	Chemistry	Two-way coupling	PM2.5	O3	Inversion	PBL	Fire	Plume	
CPCB AQI	Limited	AQI pollutants	No public evidence	Yes	Yes	No	No	No	No	No
SAMEER	Limited	AQI dissemination	No	Yes	Indirect	No	No	No	No	No
SAFAR	Yes	3-D CTM	Coupled modelling reported	Yes	Yes	Not publicly exposed	Likely internal	Fire-related capability	Not fully exposed	1–3 d
Delhi AQEWS	Yes	Coupled forecast	Research coupling	Yes	Yes	Not fully exposed	Modelled	Fire inputs	Not fully exposed	72h
CAMS	Yes	Strong global chemistry	Coupled global model	Yes	Yes	Model fields	Model fields	Fire inputs	Not plume-focused	Up to produ
HYSPPLIT	Meteorology	Optional	No	Optional	Optional	Meteorology-dependent	Meteorology-dependent	Source trajectories	Yes	User-c
FLEXPART	Meteorology	Transport/deposition	No	Optional	Optional	Meteorology-dependent	Meteorology-dependent	Strong source transport	Yes	User-c
MERRA-2	Yes	Aerosol reanalysis	Coupled assimilation framework	Aerosol	Some gases	Modelled	Modelled	Fire-related aerosol	No public local plume	Histor
FIRMS	No	No	No	No	No	No	No	Yes	No	NRT o
TROPOMI	No	Column gases	No	Indirect	Yes	No	No	Indirect	No	Daily obser
IQAir	Weather display	Aggregated data	No	Yes	Varies	No	No	No	No	Forec deper
WAQI	Limited	Aggregated data	No	Yes	Varies	No	No	No	No	Produ deper
OpenAQ	No	Measurements	No	Yes	Source-dependent	No	No	No	No	No
EDGAR	No	Emission inventory	No	Emissions	Emissions	No	No	No	No	Histor
HTAPv3.1	No	Emission inventory	No	Emissions	Emissions	No	No	No	No	Histor
GFED	No	Fire emissions	No	Emissions	Emissions	No	No	Yes	No	Histor
WRF-Chem Delhi studies	Yes	Full chemistry	Yes where configured	Yes	Yes	Yes	Yes	Configurable	Chemistry-based	Hindc
2024 northern India WRF-Chem evaluation	Yes	Aerosol/gas	Yes in model configuration	Yes	Chemistry	Modelled	Modelled	Fire emissions	Regional	Hindc

System	Weather	Chemistry	Two-way coupling	PM2.5	O3	Inversion	PBL	Fire	Plume	
2025 PHD NHM/WRF-Chem study	Yes	Chemistry	Coupled	PM2.5	Not central	Modelled	Modelled	CRB	Plume events	Hindc
2023 CNN/Transformer Delhi study	Weather features	Statistical	No	PM2.5	Unknown	No	Inputs only	Not central	No	Resea
Proposed SIH system	Yes	Physical background + ML	WRF-Chem evidence	Yes	Yes	Yes	Yes	Yes	Yes	72h

The table deliberately avoids asserting features that were not verified.

22. Thirty final USPs

#	USP	Technical value	Difficulty	Data dependency	Competitor comparison
1	Feedback-on/off evidence	Demonstrates aerosol-radiation impact	High	WRF-Chem	Strong differentiator
2	Inversion-aware forecasting	Makes trapping explicit	Medium	ERA5/WRF profiles	Strong
3	PBL collapse alert	Links mixing depth to PM accumulation	Medium	PBL + observations	Strong
4	Fire-to-NCR transport score	Avoids binary fire attribution	Medium	FIRMS + wind	Strong
5	FRP-weighted source activity	Better than hotspot counting	Medium	FIRMS FRP	Differentiator
6	Plume arrival window	Predicts likely influence timing	Medium	Trajectories	Strong
7	Pollutant-level AQI driver	Explains dominant pollutant	Low	Multi-pollutant model	Useful but not novel alone
8	Quantile forecast	Gives forecast range	Medium	Quantile model	Strong
9	Conformal calibration	Improves interval reliability	Medium	Validation residuals	Advanced
10	Station-health layer	Shows whether data are trustworthy	Low	QC metadata	Operationally valuable
11	Event replay	Reconstructs real smog cases	Low	Historical data	Excellent demo
12	Baseline skill panel	Proves improvement honestly	Low	Validation outputs	Judge credibility
13	Dynamic wind graph	Connects station relations to transport	High	Wind + station geometry	Advanced
14	Regional inflow/outflow map	Shows legal-boundary limitation	Medium	Wind/trajectory	Strong
15	Recovery-window forecast	Predicts likely improvement timing	Medium	Weather + PM	Strong
16	Rainfall-clearing module	Explains wet scavenging	Medium	Rain + PM	Useful
17	Ozone chemistry warning	Separates photochemical risk	High	O3/NOx/weather	Differentiator
18	CAMS-background blending	Corrects coarse model fields locally	Medium	CAMS + stations	Strong engineering
19	Fire confidence propagation	Carries detection uncertainty forward	Medium	FIRMS QA	Strong
20	Scenario mode	Fire/traffic perturbation comparison	High	Emissions/model	Advanced; must be labelled
21	Provenance for every pixel	Auditable outputs	Low	Metadata	Government-friendly
22	Data freshness monitor	Prevents stale forecasts	Low	Scheduler logs	Operationally valuable
23	Explanation card	Plain-language causal summary	Low	Feature attribution	Strong product feature
24	District action cards	Converts forecast to response	Medium	NCR boundaries/GRAP	Impact-oriented
25	AQI probability alerts	Better than threshold-only alert	Medium	Probabilistic forecast	Strong
26	Traffic/industrial/dust context	Adds spatial interpretation	Medium	OSM/inventory	Useful
27	Roadside-background contrast	Shows exposure heterogeneity	Medium	Station classification	Strong visual
28	Research mode/public mode	Avoids overwhelming users	Low	UX design	Strong product design
29	City-adaptor architecture	Enables expansion	Medium	Configuration files	Scalability
30	Claim guardrails	Prevents scientific overstatement	Low	Provenance/UI	Impresses technical judges

Top five for PPT

1. WRF-Chem feedback-on/off evidence.
2. Inversion and PBL-aware forecasting.
3. Fire-to-NCR plume influence.

4. Uncertainty-aware 72-hour forecast.
5. Pollutant-level explanation and decision cards.

Top three for live demonstration

1. Historical event replay.
2. Fire/plume/inversion synchronised animation.
3. Actual-versus-predicted forecast with uncertainty.

Top three for technical judges

1. Matched feedback-on/off WRF-Chem experiment.
2. Leakage-safe multi-horizon validation.
3. Fire transport feature with confidence rather than unsupported source percentage.

23. Actionable decision support

Government officials

Show:

- 72-hour AQI forecast.
- Probability of crossing 200, 300, 400 or 450.
- Dominant pollutant.
- Inversion/PBL risk.
- Fire transport confidence.
- Expected deterioration and recovery windows.
- Station reliability.
- Data freshness.

Emergency managers

Show:

- District-level risk.
- Warning lead time.
- Confidence.
- Likely duration.
- Alternative action scenarios.
- Critical transport corridors.

Schools

Show:

- Outdoor-activity risk.
- Forecast window.
- Simple recommendations.
- No diagnostic or medical claims.

Transport authorities

Show:

- Traffic-related station patterns.
- Low-ventilation periods.
- District corridors with predicted deterioration.
- Potential public-transit advisory windows.

Construction and industry

Show:

- PM10/dust risk.
- Wind-driven resuspension risk.

- High-PBL periods suitable for lower-risk activity, subject to regulatory rules.
- Forecast thresholds linked to official advisories.

GRAP integration

CAQM's GRAP framework links AQI ranges to escalating measures in Delhi-NCR. Use GRAP as a context layer, not as an autonomous enforcement engine. [\[63\]](#) [\[64\]](#) [\[65\]](#)

24. Source attribution boundaries

Defensible

- “The wind field is favourable for transport from Punjab.”
- “The trajectory ensemble intersects the NCR during the forecast window.”
- “Upwind fire activity is elevated.”
- “The fire-enhanced scenario produces higher modelled PM2.5.”
- “The local station residual is increasing relative to the regional background.”
- “Low PBL and inversion indicate accumulation risk.”

Not defensible without tagged chemistry

- “Punjab contributed 42%.”
- “Traffic caused 35%.”
- “Industry caused 20%.”
- “Dust caused exactly 25%.”
- “Stubble burning caused the entire episode.”

Recommended attribution levels

Level	Claim
Level 1	Observation: fire hotspot detected
Level 2	Transport: trajectory is favourable
Level 3	Influence: fire-derived risk score is high
Level 4	Scenario: removing/perturbing fire emissions changes modelled PM
Level 5	Source apportionment: tagged chemistry or receptor model estimate

Only use Levels 4–5 for quantitative contribution claims.

25. Uncertainty approach

Recommended implementation

Use:

1. LightGBM quantile regression for P10, P50 and P90.
2. Rolling conformal calibration on a held-out period.
3. Ensemble spread from several models or weather inputs.
4. Wider intervals at longer lead times and during unusual regimes.

Conformal prediction is useful because it can provide calibrated intervals without requiring a fully specified probability model, but calibration must be evaluated under temporal drift. [\[66\]](#)

Dashboard example

```
PM2.5: 285 µg/m³
P10-P90 range: 250–335 µg/m³
AQI: 378
Probability AQI > 300: 82%
Confidence: Medium
Main uncertainty: wind-direction forecast and fire-emission intensity
```

Do not display a narrow interval simply to make the system look accurate.

26. GIS architecture

Recommended stack

Layer	Choice	Reason
Database	PostgreSQL + PostGIS	Spatial queries, stations, grids and trajectories
Array storage	Zarr/NetCDF	Efficient gridded weather and chemistry
API	FastAPI	Python integration and rapid development
Scheduler	Celery + Redis or Prefect	Ingestion and forecast jobs
Frontend	React + TypeScript	Structured dashboard
Map	MapLibre GL JS	Open, flexible rendering
Animation	deck.gl	Fire, plume and vector layers
Lightweight fallback	Leaflet	Fast implementation
Tile generation	GeoServer or pre-generated vector/raster tiles	Dashboard delivery
Scientific tools	xarray, GeoPandas, Rasterio, GDAL	Standard geospatial stack

Map layers

- AQI heatmap.
- PM2.5 raster.
- PM10 raster.
- O3 raster.
- Station points.
- Station quality.
- FIRMS hotspots.
- FRP circles.
- Wind vectors.
- Plume trajectory.
- Arrival probability.
- NCR/district boundaries.
- PBL contour or panel.
- Inversion profile.
- Timeline slider.

Raster/vector choice

- Raster tiles for concentration fields.
- Vector tiles for stations, boundaries and fire points.
- WebGL/deck.gl for large fire/plume animation.
- Pre-rendered forecast frames for reliable demos.

27. Dashboard UX

Overview screen

Display:

- Current AQI.
- 72-hour maximum predicted AQI.
- Dominant pollutant.
- Risk category.
- Confidence.
- Forecast issue time.
- Data freshness.
- Top explanation.

Forecast screen

Display:

- Hourly PM2.5, PM10, O3 and NO2.
- P10/P50/P90.
- AQI category.
- Actual-versus-predicted mode.
- Lead-time selection.
- Station/grid toggle.

Weather screen

Display:

- Wind vectors.
- Temperature.
- Relative humidity.
- Rainfall.
- Cloud cover.
- PBL height.
- Ventilation coefficient.

Inversion screen

Display:

- Vertical temperature profile.
- Inversion base/top.
- Inversion score.
- PBL height.
- Duration.
- Trapping explanation.

Fire and plume screen

Display:

- Fire hotspots.
- FRP.
- Source-region totals.
- Trajectory ensemble.
- Arrival probability.
- Fire confidence.
- “Influence” wording, not exact causation.

Explainability screen

Display top features:

- PM2.5 at $t - 1$, $t - 6$, $t - 24$.
- PBL height.
- Wind speed/direction.
- Inversion score.
- CAMS background.
- Upwind FRP.
- Rainfall.
- Humidity.

User modes

User	Show
Official	Forecast, threshold probability, source-region influence, action timing
Scientist	Raw variables, profiles, model run metadata, metrics
Emergency manager	Alert status, lead time, duration and confidence
Public	AQI, dominant pollutant, simple forecast and safety guidance

28. Android decision

Recommendation

Build Android only as a thin alert client.

Mobile features

- Saved locations.
- AQI forecast card.
- Push alerts.
- Alert probability.
- Dominant pollutant.
- Simple uncertainty range.
- Offline last-known forecast.
- Link to full web map.

Keep on web

- WRF-Chem diagnostics.
- Fire/plume animations.
- Inversion profiles.
- Multi-layer GIS.
- Historical event analysis.
- Scientific validation.

Do not spend the core team's time reproducing the web dashboard in Kotlin.

29. Backend architecture

```
Ingestion adapters
↓
Raw object store
↓
QC workers
↓
PostGIS + Zarr
↓
Feature-generation workers
↓
Plume/physics jobs
↓
Model service
↓
AQI and uncertainty service
↓
Forecast database
↓
FastAPI
↓
React/MapLibre
```

Services

- `cpcb_ingestor`
- `weather_ingestor`
- `cams_ingestor`
- `firms_ingestor`

- `quality_control`
- `feature_builder`
- `inversion_service`
- `pbl_service`
- `trajectory_service`
- `forecast_service`
- `aqi_service`
- `uncertainty_service`
- `alert_service`
- `validation_service`
- `data_health_service`

API examples

```
GET /stations
GET /stations/{station_id}/observations
GET /forecast?lat=&lon=&horizon=72
GET /forecast/grid?pollutant=pm25&lead=
GET /aqi?lat=&lon=
GET /fires?region=&from=&to=
GET /plume?source_region=&valid_time=
GET /inversion?lat=&lon=
GET /pbl?lat=&lon=
GET /forecast/{id}/explanation
GET /validation?event=
GET /health/freshness
```

WebSocket

Use WebSockets only for:

- Forecast-run status.
- Dashboard refresh notification.
- Alert updates.

Do not use WebSockets to stream full raster data. Use cached tiles or API requests.

30. Compute and infrastructure

Prototype estimates

Component	Laptop	College server	Cloud
Tabular ML	Feasible	Easy	Easy
TCN/ST-GNN	Optional GPU	Good	Moderate cost
PostGIS	Feasible	Easy	Easy
HYSPLIT/FLEXPART	Small cases feasible	Better	Moderate
WRF-Chem	Difficult	Preferred	Expensive
Dashboard	Feasible	Easy	Easy
Multi-year gridded archive	Limited	Good storage needed	Object storage

Suggested resources

Minimum operational prototype

- 4 CPU cores.
- 8–16 GB RAM.
- 100 GB storage.
- No mandatory GPU.
- One T4-class GPU if training neural models.

WRF-Chem case

- Linux.
- 32+ CPU cores preferred.
- 64+ GB RAM desirable.
- Fast local scratch storage.
- 100–500 GB depending on domains, variables and duration.

Exact runtime cannot be promised without:

- Domain size.
- Vertical levels.
- Chemistry mechanism.
- Physics options.
- Output frequency.
- MPI configuration.
- Compiler and hardware.

Benchmark before claiming operational runtime.

Platform guidance

- Laptop: data cleaning, LightGBM, dashboard development.
- Google Colab: experiments and notebooks, not reliable operations.
- Kaggle: batch experiments, not live services.
- College server: best WRF-Chem option.
- Cloud VM: deployment and scheduled ML.
- Cloud GPU: temporary neural/WRF experiments with budget controls.

31. Honest SIH demo

Scenario

Use 6–13 November 2017 as the primary replay.

Precompute

- CPCB station observations.
- ERA5 meteorology.
- FIRMS fire detections.
- MODIS AOD.
- CAMS/MERRA-2 background.
- WRF-Chem feedback-off.
- WRF-Chem feedback-on.
- HYSPLIT/FLEXPART trajectories.
- LightGBM forecasts.
- AQI.
- Uncertainty.
- Validation metrics.
- Dashboard tile layers.

Live actions

1. Select event.
2. Show baseline weather.
3. Show fire hotspots and FRP.
4. Show wind and plume transport.
5. Show inversion and PBL collapse.
6. Show pollutant forecasts.
7. Show AQI driver.
8. Toggle feedback-on/off.

- 9. Compare actual and predicted values.
- 10. Display forecast interval.
- 11. Trigger deterioration alert.
- 12. Run a clearly labelled fire-reduction scenario.

What can run live

- Map layer toggles.
- Time slider.
- Feature explanation.
- AQI recomputation from cached pollutant forecast.
- Alert threshold selection.
- Scenario visualisation using precomputed perturbation outputs.

What should not run live unless tested

- Full WRF-Chem.
- Large-area HYSPLIT/FLEXPART ensemble.
- Downloading external data during the pitch.
- Uncached satellite ingestion.

32. Claims judges may challenge

Weak claim	Replacement
“100% accurate”	“Validated against persistence and ML baselines by lead time and event regime”
“Exact source attribution”	“Modelled source-region influence under documented assumptions”
“Real-time WRF-Chem”	“Hybrid near-real-time forecast with offline WRF-Chem coupling evidence”
“Street-level”	“Approximately 5-km grid plus station correction”
“Fire hotspot equals PM2.5 emission”	“Fire detection and FRP-derived activity proxy”
“Stubble burning caused 40%”	“Fire-emission sensitivity changed modelled PM2.5 by X in this experiment”
“Observed PBL”	“Reanalysis/model-derived PBL height”
“Measured inversion everywhere”	“Profile-based inversion proxy calibrated against available observations”
“Replaces SAFAR”	“Complementary explainable decision-support layer”
“Works nationwide”	“Reusable architecture requiring city-specific calibration”
“Satellite observes all fires”	“Satellite fire detections have confidence, cloud and overpass limitations”
“AI models chemistry”	“AI corrects physical/background forecasts using chemistry-aware features”

33. Feasibility score

Dimension	Score
Data availability	7/10
Scientific feasibility	8/10
Engineering feasibility	8/10
ML feasibility	9/10
WRF-Chem feasibility	5/10
Compute feasibility	6/10
Real-time feasibility	7/10
Demo feasibility	9/10
SIH compliance	8/10
Innovation	8/10
Impact	9/10
Scalability	8/10
Deployment feasibility	7/10

Overall

7.5/10 for the hybrid architecture.

The concept is feasible if scope is controlled. WRF-Chem must be treated as a research and compliance layer, not as the only operational forecast engine.

34. Risk register

Risk	Probability	Impact	Detection	Mitigation	Backup
CPCB history unavailable	High	High	Historical download test	Use cached period, OpenAQ/OGD, documented subset	CAMS + available stations
Portal/API change	Medium	High	Health checks	Adapter architecture and snapshots	Cached data
WRF-Chem failure	High	High	Early smoke test	Start early; use known configuration	WRF/ML hybrid
Insufficient compute	High	High	Runtime benchmark	Coarse domain and one event	Physics-guided ML
Emission uncertainty	High	High	Sensitivity experiments	Use priors and uncertainty	Influence score
FIRMS cloud gaps	Medium	Medium	QA flags	Fuse MODIS/VIIRS and multi-day windows	Reanalysis fire product
Station failure	Medium	High	Flatline/completeness checks	Station-health filtering	Background/residual model
Model drift	Medium	Medium	Rolling validation	Seasonal recalibration	Persistence fallback
Weak 72-hour accuracy	Medium	High	Lead-time metrics	Ensemble, uncertainty, regime features	Alert probability
API outage	Medium	High	Monitoring	Cache demo and last valid forecast	Local replay
Licence violation	Medium	High	Terms audit	Attribution and controlled redistribution	Replace source
Overclaiming coupling	High	High	Technical review	Feedback-on/off evidence	Honest hybrid wording
TROPOMI cloud gaps	Medium	Low-medium	QA flags	Optional feature, not dependency	CAMS/MERRA-2
Map performance	Medium	Medium	Browser profiling	Vector tiles and caching	Static raster frames
AQI implementation error	Medium	High	Compare CPCB calculator	Versioned breakpoint tests	Use official calculator logic

35. Development roadmap

Phase	Deliverables	Difficulty	Approximate time
1. Data acquisition	Source registry, terms, sample downloads	Medium	3-5 days
2. Data engineering	Hourly canonical dataset, QC	High	1-2 weeks
3. Baselines	Persistence, climatology, LightGBM	Medium	4-7 days
4. Weather/pollution model	Multi-pollutant 72-hour outputs	High	1-2 weeks
5. Fire/plume	FIRMS features, trajectories, maps	High	1 week
6. WRF-Chem	One event and feedback comparison	Very high	2-4 weeks
7. Hybrid AI	Physical background and residual correction	High	1 week
8. Validation	Metrics, event/regime analysis	Medium	4-7 days
9. GIS	PostGIS, tiles, animations	High	1 week
10. Dashboard	Main screens and API integration	High	1-2 weeks
11. Deployment	Docker, scheduler, monitoring	Medium	4-7 days
12. PPT/demo	Evidence, replay and judge Q&A	Medium	3-5 days

Critical sequencing

1. Build the data pipeline first.
2. Start WRF-Chem compilation immediately in parallel.
3. Finish persistence and LightGBM baselines before deep learning.
4. Build the historical replay before the live dashboard.
5. Validate before claiming USPs.
6. Freeze scientific claims before PPT design.

36. Six-person allocation

Person	Role	Primary output
1	ML/AI	Baselines, multi-horizon model, uncertainty, explainability
2	Atmospheric modelling	WRF-Chem, PBL, inversion, chemistry and scientific defence
3	Data engineering	CPCB, ERA5, CAMS, FIRMS ingestion, QC and provenance
4	Backend/MLOps	FastAPI, database, scheduler, Docker and monitoring
5	GIS/frontend	PostGIS, MapLibre, dashboard and animation
6	Android/product/PPT	Mobile alerts, personas, demo flow, slides and Q&A

Dependencies:

- Person 2 defines valid scientific features.
- Person 3 provides leakage-safe data.
- Person 1 creates models and validation.
- Person 4 exposes stable APIs.
- Person 5 consumes APIs and renders maps.
- Person 6 translates validated capabilities into user and judge language.

37. Final technology stack

Data

- Python.
- pandas/Polars.
- xarray.
- NetCDF4.
- Zarr.
- Dask only where needed.

Scientific

- WRF-Chem.
- WRF/WPS.
- HYSPLIT or FLEXPART.
- MetPy.
- GeoPandas.
- Rasterio.
- GDAL.

ML

- LightGBM.
- XGBoost.
- scikit-learn.
- PyTorch for TCN/ST-GNN.
- SHAP or native model explanations.
- Quantile regression.
- Conformal calibration.

Backend

- FastAPI.
- PostgreSQL/PostGIS.
- Redis.
- Celery or Prefect.
- Docker.
- Nginx.

Frontend/GIS

- React.
- TypeScript.
- MapLibre GL JS.
- [deck.gl](#).
- Leaflet fallback.

Android

- Kotlin.
- WorkManager.
- Firebase Cloud Messaging if allowed and appropriate.
- Cached REST responses.

Monitoring

- Prometheus/Grafana if infrastructure permits.
- Otherwise structured logs plus `/health` endpoints.
- Model/data freshness dashboard.

38. Cost

Prototype estimate

Component	Approximate cost
Open-source software	₹0
Public scientific data	Usually ₹0, subject to registration/terms
Local development	₹0 if laptops exist
Small cloud API	₹0–₹2,000/month
Storage/database	₹0–₹3,000/month
Temporary GPU	₹0–₹10,000
WRF-Chem cloud compute	Potentially much higher
Domain	Approximately ₹800–₹1,500/year
Map services	₹0 with self-hosted/open strategy; otherwise usage-dependent
Android prototype	₹0 development cost

The main financial risk is repeated WRF-Chem cloud execution, not the dashboard.

39. Impact

Government

- Earlier preparation for pollution episodes.
- Better coordination between Delhi, Haryana, UP and Rajasthan.
- More transparent basis for issuing advisories.
- Regional transport context beyond Delhi's administrative boundary.

Disaster management

- 72-hour deterioration window.
- Probability-based alerts.
- Expected recovery window.
- Fire/plume transport risk.

Schools and public health

- Outdoor-activity timing.
- Simple air-quality alerts.
- Better communication of uncertainty.

Do not claim that the system itself will save a specific number of lives. The defensible claim is that earlier and more interpretable forecasts can support earlier interventions.

Agriculture

- Fire activity monitoring.
- Source-region trend visualisation.
- Scenario analysis of reduced burning.
- No blame assignment from a hotspot alone.

Industry, construction and transport

- PM10/dust risk.
- Low-ventilation warnings.
- Forecast support for compliance and scheduling.
- Integration with official regulatory frameworks.

40. Scalability

Build city-specific configuration rather than hard-coded Delhi logic.

```
city: delhi_ncr
bbox: [...]
grid_resolution: 0.05
stations: [...]
source_regions: [Punjab, Haryana, UP, Rajasthan]
aqi_standard: CPCB
seasonal_regimes: [...]
```

Reusable:

- Ingestion adapters.
- QC.
- Feature store.
- Forecast API.
- AQI service.
- Alert engine.
- Uncertainty.
- GIS frontend.
- Validation framework.

City-specific:

- Emissions.
- Source regions.
- Station network.
- Chemical regime.
- Meteorology.
- Calibration.
- Pollutant relevance.

Mumbai, Pune, Ahmedabad, Lucknow, Kolkata, Bengaluru and Hyderabad require new calibration and source-region definitions.

41. Literature priority list

Most important for your design

1. WRF-Chem evaluation over northern India

Title: Evaluation of WRF-Chem-simulated meteorology and aerosols over northern India during September–November 2016.

Year: 2024.

Journal: Atmospheric Chemistry and Physics.

Use: Shows realistic meteorology/aerosol capability and important peak-PM2.5 biases. ^[62] [6]

2. Improving regional air-quality predictions in the Indo-Gangetic Plain

Year: 2021.

Journal: Atmospheric Chemistry and Physics.

Use: Fire activity, FRP approximation, WRF-Chem, aerosol-radiation feedback and Delhi-relevant transport. It reports that fire aerosols reduced surface radiation, altered PBL height and increased aerosol concentrations in the model. [\[7\]](#)

3. Crop-residue burning forecast/data assimilation study

Authors: Kumar and collaborators.

Use: 72-hour Delhi forecasting, MODIS AOD assimilation, fire emissions and aerosol-radiation feedback. [\[5\]](#)

4. Stubble-burning activities in north-western India in 2021

Year: 2023.

Journal: Heliyon.

Use: MODIS fire counts, Punjab/Haryana burning and modelled CO/PM_{2.5} contribution. [\[68\]](#) [\[34\]](#)

5. Post-monsoon crop-residue-burning impact over Punjab–Haryana–Delhi

Year: 2025.

Journal: Atmospheric Chemistry and Physics.

Use: NHM/WRF-Chem and dense observations; estimated average CRB contribution around 30–34%, increasing to approximately 50–56% during plume events. These values are study-specific, not universal. [\[69\]](#)

6. Aerosol radiative effects on Delhi PM_{2.5} and O₃

Year: 2024.

Use: Demonstrates that aerosol direct and photolysis effects can alter surface O₃ and PM_{2.5} differently. [\[70\]](#)

7. November 2016 severe pollution study

Use: Historical event selection and severe PM_{2.5} evidence. [\[52\]](#)

8. Great Smog of Delhi

Use: Historical event narrative and reported extreme PM_{2.5}/AQI values. [\[50\]](#)

9. Delhi early-warning system

Use: Prior-art benchmark; establishes that 72-hour Delhi forecasting already exists. [\[60\]](#) [\[62\]](#)

10. CPCB AQI methodology

Use: Official AQI formula, pollutants, averaging windows and breakpoints. [\[46\]](#) [\[47\]](#) [\[71\]](#)

11. ERA5 documentation

Use: Historical weather, pressure-level profiles and PBL features. [\[12\]](#) [\[13\]](#)

12. CAMS documentation

Use: Chemical background and forecast products. [\[72\]](#) [\[73\]](#) [\[14\]](#)

13. FIRMS documentation

Use: Fire confidence, FRP and active-fire limitations. [\[19\]](#) [\[20\]](#) [\[21\]](#)

14. HYSPLIT/FLEXPART documentation

Use: Plume transport design and licensing. [\[36\]](#) [\[38\]](#) [\[39\]](#)

15. Air-quality ML literature

Use ML papers only for methodological comparison. Do not transfer their headline R^2 , RMSE or MAE to your project without reproducing the dataset and protocol. A 2025 review and recent studies report strong XGBoost/ensemble performance in some settings, but results vary by city, horizon, leakage controls and data quality. [\[44\]](#) [\[45\]](#)

42. Repositories and official software

Software	URL	Use	Licence/access	Feasibility
WRF	official WRF repository	Weather and WRF-Chem foundation	Public domain according to repository	Feasible with Linux
WRF-Chem	NCAR WRF-Chem page	Coupled chemistry	Official documentation/download	Difficult but required for one case
HYSPLIT	NOAA HYSPLIT	Trajectories/dispersion	Use agreement applies	Good for demo
FLEXPART	official development mirror	Lagrangian transport	GPLv3	Good advanced option
xarray	xarray	NetCDF/GRIB processing	Open source	Essential
GeoPandas	GeoPandas	Vector GIS	Open source	Essential
Rasterio	Rasterio	Raster processing	Open source	Useful
PostGIS	PostGIS	Spatial database	Open source	Recommended
MapLibre	MapLibre GL JS	Web maps	Open source	Recommended
deck.gl	deck.gl	WebGL visualisation	Open source	Useful for plume/fire animation

The official WRF repository describes WRF as open-source public-domain code with unrestricted use; FLEXPART's repository identifies GPL-3.0 licensing. ^[43] ^[41]

43. Exact bookmark list

Must use

- [CPCB real-time air-quality data](#) — observations and station monitoring. ^[9]
- [CPCB NCRAQI](#) — NCR station and AQI access. ^[10]
- [CPCB CAAQMS list](#) — station names, coordinates and agencies. ^[8]
- [CPCB AQI method](#) — calculation method. ^[47]
- [ERA5 CDS](#) — historical weather. ^[13]
- [NASA FIRMS](#) — fire observations. ^[19]
- [CAMS global forecasts](#) — chemical background. ^[73]
- [WRF-Chem](#) — coupled model. ^[74]
- [WRF source](#) — model repository. ^[43]
- [NOAA HYSPLIT](#) — transport. ^[36]
- [NCRPB constituent areas](#) — official NCR boundary. ^[2]

Strongly recommended

- MERRA-2.
- MODIS AOD.
- TROPOMI/Sentinel-5P.
- GFS.
- FLEXPART.
- HTAPv3.1.
- GFED.
- OSM.
- Current CPCB/DPCC/HSPCB/UPPCB registry.
- Radiosonde or boundary-layer observations.

Optional

- IMD/NCMRWF detailed forecast data.
- AERONET.
- GFAS/FINN.
- Land-use and population grids.
- Industrial and transport inventories.
- INSAT/MOSDAC products.

Research only

- Multi-year 1-km WRF-Chem.
- Full chemical data assimilation.
- Tagged-tracer national source apportionment.
- Neural ODE/PINN.
- Fully automated policy optimisation.

Not worth initial effort

- Full Android replica.
- Custom satellite fire detector.
- Nationwide operations.
- Street-level forecasts.
- Exact source attribution.
- Transformer before tabular baselines.
- Live WRF-Chem without a tested pipeline.

44. Final architecture

```
CPCB/DPCC/HSPCB/UPPCB/RSPCB observations
ERA5 historical weather
GFS forecast weather
CAMS/MERRA-2 atmospheric composition
NASA FIRMS VIIRS/MODIS fires
MODIS AOD and TROPOMI
NCRPB boundaries and OSM
↓
Source adapters + scheduled ingestion
↓
Immutable raw storage + provenance
↓
Quality control + station health
↓
UTC/IST alignment + unit normalisation
↓
Station/background spatio-temporal fusion
↓
ERA5/GFS/WRF inversion and PBL diagnostics
↓
FIRMS FRP + source-region fire features
↓
HYSPLIT/FLEXPART transport influence
↓
WRF-Chem historical feedback experiment
↓
LightGBM multi-pollutant residual correction
↓
Quantile + conformal uncertainty
↓
PM2.5, PM10, O3, NO2 forecasts
↓
CPCB AQI engine
↓
PostGIS + Zarr forecast storage
↓
MapLibre/deck.gl GIS
↓
React dashboard
↓
Kotlin alerts and mobile summary
```

45. Product priorities

P0

- Current CPCB station observations.
- 72-hour PM2.5 forecast.
- PM10 and O3 forecast.
- NO2 where data permit.
- CPCB AQI.

- Delhi-NCR map.
- Weather layer.
- PBL and inversion diagnostics.
- FIRMS fire map.
- Plume trajectory.
- Uncertainty range.
- Historical replay.
- Actual-versus-predicted validation.
- Data freshness labels.

P1

- WRF-Chem feedback-on/off evidence.
- Fire-to-NCR transport score.
- Pollutant-driver explanation.
- Recovery-window forecast.
- Station health.
- Regime-specific metrics.
- Probability threshold alerts.
- District decision cards.

P2

- Tagged source attribution.
- Fire-reduction scenario.
- Traffic/construction scenario.
- ST-GNN.
- Multi-model weather ensemble.
- TROPOMI assimilation.
- Automated anomaly detection.

P3

- Full operational WRF-Chem.
- Multi-city expansion.
- Chemical data assimilation.
- Exposure-route planning.
- Policy optimisation.
- Public API.

46. PPT plan

Slide 1 – The problem

Message: Delhi-NCR pollution is a coupled weather–chemistry disaster-management problem.

Include:

- SIH26082 title.
- 72-hour requirement.
- Delhi-NCR map.
- One severe-event visual.

Slide 2 – Why existing forecasts are insufficient for this use case

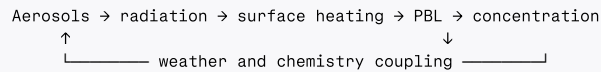
Show:

- Observations.
- Generic ML.
- Coarse chemical forecasts.
- Proposed integrated layer.

Do not say government systems do not exist.

Slide 3 — Scientific feedback loop

Show:



Slide 4 — Delhi-NCR source geography

Show:

- Legal NCR.
- Punjab/Haryana/UP/Rajasthan.
- Source-to-receptor arrows.
- Monitoring stations.

Slide 5 — Data fusion

Show:

- CPCB.
- ERA5/GFS.
- CAMS.
- FIRMS.
- MODIS/TROPOMI.
- OSM/NCRPB.

Slide 6 — Coupled scientific layer

Show:

- WRF-Chem.
- Feedback-on/off.
- PBL, radiation, chemistry and emissions.

Slide 7 — Fire/plume module

Show:

- FIRMS hotspots.
- FRP.
- Wind field.
- Trajectory ensemble.
- Arrival probability.

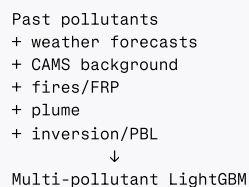
Slide 8 — Inversion and PBL

Show:

- Vertical temperature profile.
- Inversion layer.
- PBL height.
- PM2.5 time series.

Slide 9 — Hybrid AI

Show:



↓
Quantile uncertainty

Slide 10 — AQI explanation

Show:

- PM2.5, PM10 and O3 forecast.
- CPCB sub-indices.
- Dominant pollutant.
- AQI category.

Slide 11 — Dashboard

Show your own screenshot:

- Map.
- Timeline.
- Fire layer.
- Inversion/PBL panel.
- Confidence range.

Slide 12 — Validation

Show:

- Persistence versus proposed.
- Lead-time error.
- Historical event replay.
- Station holdout.

Slide 13 — Top five USPs

Use:

1. Coupling evidence.
2. Inversion/PBL.
3. Fire-to-NCR transport.
4. Uncertainty.
5. Decision-support explanation.

Slide 14 — Impact

Map features to:

- Officials.
- Emergency managers.
- Schools.
- Transport.
- Construction.
- Agriculture.

Slide 15 — Feasibility and roadmap

Show:

- P0.
- WRF-Chem case.
- Hybrid operational system.
- Risks and mitigations.
- Expansion path.

47. Visual asset plan

Visual	Build/source
NCR map	Generate from NCRPB boundary
Station map	Generate from current CPCB registry
Fire hotspot map	Generate from FIRMS with attribution
Plume animation	Generate from HYSPLIT/FLEXPART
Inversion diagram	Create in PowerPoint/Figma
PBL diagram	Generate from ERA5/WRF profiles
AQI heatmap	Generate from your forecast grid
Actual-predicted graph	Generate from CPCB observations
Feedback comparison	Generate from WRF-Chem runs
Satellite event image	Use NASA/ISRO/ESA product with attribution and terms check
Dashboard screenshot	Create yourself
Architecture diagram	Create yourself
Forecast uncertainty chart	Create from quantile model

The search results include a NASA satellite visual of the 5 November 2016 Delhi pollution event, but for the PPT it is safer to use a properly attributed source image or recreate the visual from data.

48. Judge Q&A

Scientific

1. What is two-way coupling?

Weather transports and mixes pollutants, while aerosols affect radiation, photolysis and boundary-layer dynamics. We demonstrate this through matched WRF-Chem feedback-on/off runs and use the results to inform the operational hybrid model. ^[7]

2. Why does PBL matter?

It approximates the vertical mixing volume. A shallow PBL generally increases surface concentrations for similar emissions.

3. Why is Delhi not a closed box?

Regional transport from Punjab, Haryana, UP, Rajasthan and the wider Indo-Gangetic Plain can enter or leave NCR.

4. Is an inversion the same as low PBL?

No. They are related but distinct diagnostics. Inversion describes the vertical temperature structure; PBL height describes turbulent mixing depth.

5. Is FIRMS fire activity equal to emissions?

No. FIRMS provides active-fire detections and FRP. Emission mass requires fuel, burned area, combustion completeness and emission factors.

6. Why include ozone?

Ozone is an AQI pollutant with different chemistry and averaging requirements from PM.

7. Why predict pollutants before AQI?

AQI is derived from pollutant concentrations. Pollutant forecasts preserve chemical meaning and support explanations.

8. What is secondary aerosol?

Particle material formed in the atmosphere from gas precursors such as NO_x, SO₂, NH₃, VOCs and oxidants.

9. What is the limitation of your inversion index?

It is an operational trapping-risk index, not a universal regulatory or observational standard.

10. What proves your model is scientifically meaningful?

Event-based validation, station holdout, lead-time skill, feedback-on/off experiments and ablations of fire/inversion/PBL features.

ML

11. Why LightGBM?

The inputs are heterogeneous tabular features and LightGBM is fast, strong and practical for quantile forecasting.

12. Why not Transformer?

We will use it only if it beats leakage-safe baselines. Architecture popularity is not a substitute for validation.

13. How do you avoid leakage?

Every feature is associated with an availability time. We use blocked temporal splits and station holdouts.

14. What is your baseline?

Persistence, seasonal average, weather-only ML and pollution-weather ML.

15. What if your model is worse at 72 hours?

We report that honestly, widen uncertainty and use probability-based warnings rather than hiding the weakness.

16. How will you handle severe episodes?

Use event-stratified training/evaluation, regime features, sampling strategy and severe-event recall.

17. How do you forecast unmonitored grid cells?

Use a physical/background field plus station-residual correction and display spatial uncertainty.

18. How do you explain a prediction?

Use feature contributions from lags, wind, PBL, inversion, CAMS background, fire FRP and rainfall.

19. How do you calibrate uncertainty?

Use quantile regression and conformal calibration on a temporally held-out period.

20. Why multi-output prediction?

Pollutants interact and the output is easier to validate consistently, while still allowing separate pollutant losses.

WRF-Chem

21. Are you running WRF-Chem operationally?

Not as the sole operational path. We use WRF-Chem for an actual coupled historical experiment and a faster hybrid model for operational forecasting.

22. Why not run it live?

A reliable live cycle needs tested emissions preprocessing, chemistry initialisation, compute, monitoring and recovery. We will not claim that without evidence.

23. What is your fallback?

GFS/CAMS/CPCB/FIRMS plus plume trajectories and ML residual correction.

24. Which emissions do you use?

Anthropogenic inventory, fire-derived emissions or FRP proxy, dust and available biogenic emissions, with configuration and sensitivity documented.

25. What is your feedback test?

Matched simulations differing only in aerosol-radiation/related feedback settings, comparing radiation, temperature, PBL and pollution.

26. What causes WRF-Chem error?

Meteorology, emissions, chemistry, plume rise, boundary conditions, land surface and numerical configuration.

27. What chemistry mechanism?

A supported gas/aerosol configuration selected after smoke testing and sensitivity evaluation; the exact version and namelist will be disclosed.

28. How do you validate PBL?

Compare with available radiosonde, lidar/sodar or other boundary-layer observations, and test its relationship with observed PM.

29. How do you validate plume transport?

Compare trajectory pathways with wind fields, fire-region timing, satellite smoke evidence and observed downwind concentration changes.

30. Why not use only HYSPLIT?

HYSPLIT helps with transport, but it does not by itself represent full two-way aerosol-meteorology feedback.

Data

31. Is CPCB data open?

Current data and station resources are publicly exposed, but historical bulk access and endpoint stability must be verified operationally. ^[9] ^[8]

32. Why ERA5?

It provides long, consistent historical meteorology and vertical profiles for training and inversion analysis. ^[13]

33. Why GFS?

It is suitable as an operational forecast-weather source, unlike retrospective ERA5.

34. Why CAMS?

It provides chemical background fields and multiple gases/aerosols at regional scale, but not street-scale truth. ^[14]

35. Why use satellites?

They provide regional spatial context unavailable from sparse stations, while ground stations provide surface measurements.

36. How do you handle satellite gaps?

Use quality flags and treat satellite fields as optional/complementary features.

37. How do you handle station duplicates?

Use station code and coordinates, preserve registry version and flag duplicate site names.

38. What is the current station count?

The accessible CPCB snapshot lists 358 CAAQMS stations nationally and 39 in Delhi, but the live count changes. ^[8]

39. Why do station counts differ?

Sources may count active, registered, historical, relocated or agency-specific stations differently.

40. Can you commercially deploy all data?

Not automatically. Each source must be checked separately; OSM, GADM, HYSPLIT and government portals have different terms. ^[38] ^[31] ^[27]

Accuracy and deployment

41. What accuracy do you guarantee?

None in advance. We will publish actual lead-time and event-wise metrics.

42. What if CPCB fails?

Use cached data and fallback products, and visibly mark freshness.

43. What if the weather forecast is wrong?

Use an ensemble or multiple forecast runs and let uncertainty widen.

44. What if the model drifts?

Monitor rolling error and recalibrate seasonally.

45. How much storage is required?

Approximately 100–300 GB for an initial station, weather and selected event archive; WRF-Chem archives can require considerably more.

46. Can this run on a laptop?

The ML and dashboard can. Full WRF-Chem is better on a Linux server.

47. What is live and what is precomputed?

The API, map controls, alert logic and cached forecasts can be live; WRF-Chem event runs should be precomputed unless operational runtime is proven.

48. How do you protect the API?

Use authentication for administrative endpoints, rate limiting, input validation, secrets management and read-only public forecast endpoints.

49. How do you protect data provenance?

Store source URL, retrieval time, version, licence, processing code version and model-run identifier.

50. What is the minimum credible product?

CPCB + weather + CAMS + FIRMS + inversion/PBL + pollutant-level LightGBM + AQI + uncertainty + plume map + one WRF-Chem feedback experiment.

Fake-coupling traps

51. If weather enters your ML model, why is that not two-way coupling?

Because the ML model does not feed predicted chemical changes back into radiation, temperature or PBL equations.

52. If PM2.5 affects the ML-predicted PBL feature, is that coupling?

Not physical two-way coupling unless the feedback is based on a validated model relationship and is explicitly distinguished as statistical feedback.

53. Is a WRF-Chem screenshot enough?

No. We must show configuration, emissions, feedback settings, runtime, output and validation.

54. Is CAMS coupled?

CAMS is a coupled atmospheric-composition forecasting system, but using CAMS outputs as features in our model does not mean our own model is two-way coupled.

55. Is HYSPLIT plus an ML correction two-way coupled?

No. It is a transport-plus-statistical hybrid unless chemical variables modify meteorology.

56. What result would disprove your feedback claim?

If feedback-on and feedback-off simulations produce no material difference, or if the difference is not attributable to the intended feedback process, we would not claim operationally meaningful two-way coupling.

57. How do you prevent feedback from being manually fabricated?

Use matched experiments, preserve namelists, archive output and report the differences.

58. Can your operational model claim two-way coupling?

Only the WRF-Chem component should be called genuinely two-way coupled. The operational ML layer should be called physics-informed or WRF-Chem-informed.

49. Final executive verdict

If I were evaluating this SIH proposal, would I build it?

MODIFY, then GO.

Strongest reason

It addresses a high-impact disaster-management problem with a technically rich and visually compelling solution.

Biggest technical risk

WRF-Chem setup, emissions preprocessing and compute.

Biggest data risk

Stable historical and real-time access to CPCB and institutional datasets.

Biggest competitive advantage

The system explains the forecast through:

- Pollutant concentration.
- Weather.
- PBL.
- Inversion.
- Fire activity.
- Plume transport.
- Uncertainty.
- Action timing.

Minimum viable scientifically defensible solution

- CPCB station observations.
- ERA5 historical meteorology.
- GFS forecast meteorology.
- CAMS chemical background.
- FIRMS fires and FRP.
- Inversion/PBL diagnostic.
- Multi-pollutant LightGBM.
- CPCB AQI engine.
- Quantile/conformal uncertainty.
- HYSPLIT/FLEXPART plume layer.
- One reproducible WRF-Chem feedback-on/off case.
- Event-based validation.
- GIS dashboard.

Ideal solution

- Operational WRF-Chem.
- Tagged tracers.
- Chemical data assimilation.
- High-resolution nested domain.
- Multi-model weather and chemistry ensemble.
- Calibrated uncertainty.
- Government workflow integration.
- Transferable city adapters.

Demonstrate

- One historic severe event.
- Actual versus predicted concentrations.
- Fire/plume movement.
- Inversion/PBL evolution.
- WRF-Chem feedback difference.
- AQI explanation.
- Uncertainty.
- Decision alert.
- Fire-reduction scenario labelled as a modelled scenario.

Do not attempt

- Full national operational WRF-Chem.
- Street-level claims.
- Exact source percentages.
- Full Android clone.
- Neural architecture before baselines.
- Unvalidated policy optimisation.
- Any claim that the system replaces CPCB or SAFAR.

BUILD THIS checklist

Datasets

- ☐ CPCB CAAQMS station registry.
- ☐ CPCB/DPCC/HSPCB/UPPCB/RSPCB observations.
- ☐ CPCB AQI breakpoints and calculator.
- ☐ ERA5 single-level hourly data.
- ☐ ERA5 pressure-level profiles.
- ☐ GFS forecast weather.
- ☐ CAMS composition forecast.
- ☐ MERRA-2 aerosol data.
- ☐ FIRMS VIIRS fire data.
- ☐ FIRMS MODIS fire data.
- ☐ Fire confidence and FRP.
- ☐ MODIS AOD for the selected event.
- ☐ TROPOMI NO₂/CO/O₃ where practical.
- ☐ NCRPB boundaries.
- ☐ District boundaries.
- ☐ OSM road/industrial features.
- ☐ HTAPv3.1 or EDGAR emissions.
- ☐ GFED/GFAS/FINN fire emissions if WRF-Chem permits.
- ☐ One severe event archive.

Models

- ☐ Persistence.
- ☐ Historical mean.
- ☐ Weather-only LightGBM.
- ☐ Pollution-weather LightGBM.
- ☐ Fire/inversion/PBL LightGBM.
- ☐ Quantile LightGBM.
- ☐ Conformal calibration.
- ☐ Optional TCN/ST-GNN.
- ☐ HYSPLIT or FLEXPART.
- ☐ WRF-Chem feedback-off.
- ☐ WRF-Chem feedback-on.
- ☐ Fire-emission sensitivity.

Software

- ☐ Python.
- ☐ pandas/Polars.
- ☐ xarray.
- ☐ NetCDF/Zarr.
- ☐ GeoPandas.
- ☐ Rasterio/GDAL.
- ☐ LightGBM/XGBoost.
- ☐ PyTorch.
- ☐ WRF/WPS/WRF-Chem.
- ☐ HYSPLIT/FLEXPART.
- ☐ PostgreSQL/PostGIS.
- ☐ FastAPI.
- ☐ Redis/Celery or Prefect.
- ☐ Docker.
- ☐ React/TypeScript.
- ☐ MapLibre/deck.gl.
- ☐ Kotlin Android alerts.

Modules

- ☐ Ingestion adapters.
- ☐ Data-provenance registry.
- ☐ Station quality control.
- ☐ Sensor-health monitoring.
- ☐ Timestamp/unit normalisation.
- ☐ Background/residual spatial fusion.
- ☐ Inversion index.
- ☐ PBL and ventilation features.
- ☐ FIRMS fire aggregation.
- ☐ FRP-weighted source score.
- ☐ Trajectory/plume module.
- ☐ WRF-Chem research module.
- ☐ ML forecast service.
- ☐ AQI engine.
- ☐ Quantile/conformal uncertainty.
- ☐ Validation service.
- ☐ Alert engine.
- ☐ Data freshness monitor.

Screens

- ☐ Overview.
- ☐ Live AQI.
- ☐ 72-hour forecast.
- ☐ PM2.5 map.
- ☐ PM10 map.
- ☐ O3 map.
- ☐ NO2 map.
- ☐ Station map.
- ☐ Weather map.
- ☐ Inversion/PBL panel.
- ☐ Fire map.
- ☐ Plume map.
- ☐ Explainability panel.
- ☐ Uncertainty panel.
- ☐ Historical replay.
- ☐ Alerts.
- ☐ Station details.
- ☐ Scientific/model metadata.

Experiments

- ☐ Persistence comparison.
- ☐ Historical-average comparison.
- ☐ Weather-only ablation.
- ☐ Fire-feature ablation.
- ☐ Inversion-feature ablation.
- ☐ PBL-feature ablation.
- ☐ CAMS-background ablation.
- ☐ Spatial station holdout.
- ☐ Temporal holdout.
- ☐ Severe-smog evaluation.
- ☐ Fire-event evaluation.
- ☐ Rainfall-clearing evaluation.
- ☐ WRF-Chem feedback-on/off.
- ☐ Fire-emission sensitivity.
- ☐ Prediction interval calibration.

Graphs

- ☐ Actual versus predicted PM2.5.
- ☐ Actual versus predicted PM10.
- ☐ Actual versus predicted O3.
- ☐ Lead-time MAE/RMSE.
- ☐ Skill versus persistence.
- ☐ Station-wise error map.
- ☐ AQI-category confusion matrix.
- ☐ Inversion/PBL versus PM2.5.
- ☐ Fire FRP versus PM2.5.
- ☐ Feedback-on/off comparison.
- ☐ Uncertainty bands.
- ☐ Alert calibration.
- ☐ Recovery-window chart.

Maps

- [] Official NCR boundary.
- [] Complete current station map.
- [] Source-region map.
- [] Fire hotspot map.
- [] FRP intensity map.
- [] Wind vector map.
- [] Plume trajectory map.
- [] Arrival-probability map.
- [] AQI heatmap.
- [] Pollutant-specific heatmaps.
- [] PBL/inversion map or panel.
- [] 72-hour forecast animation.

PPT assets

- [] Verified official PS document.
- [] NCR geography map.
- [] Scientific feedback diagram.
- [] Source-to-receptor diagram.
- [] Data architecture.
- [] WRF-Chem domain diagram.
- [] Feedback-on/off evidence.
- [] Fire/plume visual.
- [] Inversion/PBL figure.
- [] ML pipeline.
- [] Dashboard screenshot.
- [] Validation chart.
- [] Uncertainty chart.
- [] Top-five USP slide.
- [] Feasibility/risk slide.
- [] 50-question Q&A sheet.
- [] Source/licence register.

✱✱

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